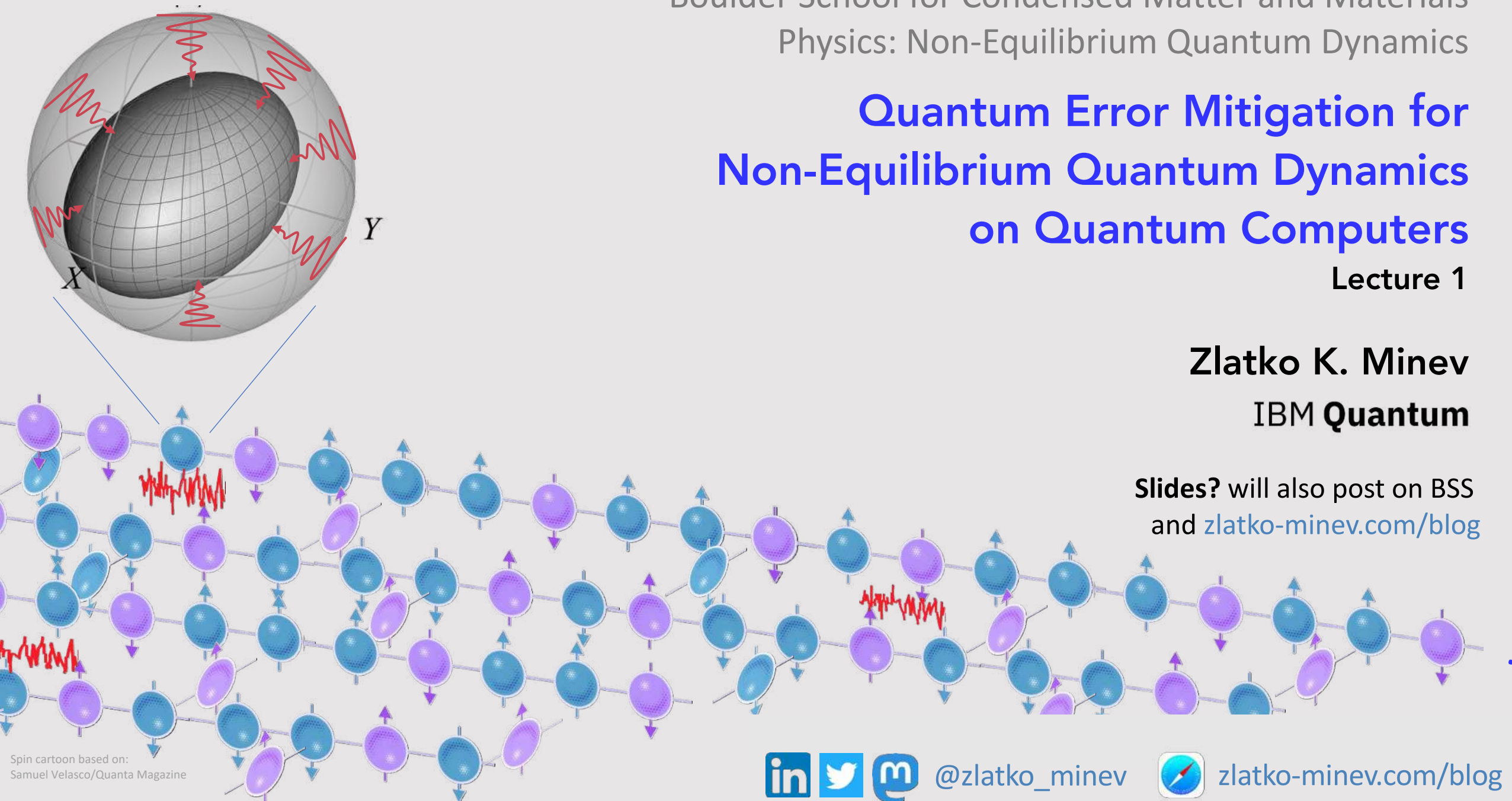


Quantum Error Mitigation for Non-Equilibrium Quantum Dynamics on Quantum Computers

Lecture 1

Zlatko K. Mineev
IBM Quantum

Slides? will also post on BSS
and zlatko-minev.com/blog



Where can you find things?

Lecture Slides

Boulder School for Condensed Matter and Materials Physics

boulderschool.yale.edu/2023/boulder-school-2023-lecture-notes



Will also post on zlatko-minev.com/education

Tutorials and additional lecture notes

Twirling, Measurements and Walsh-Hadamard

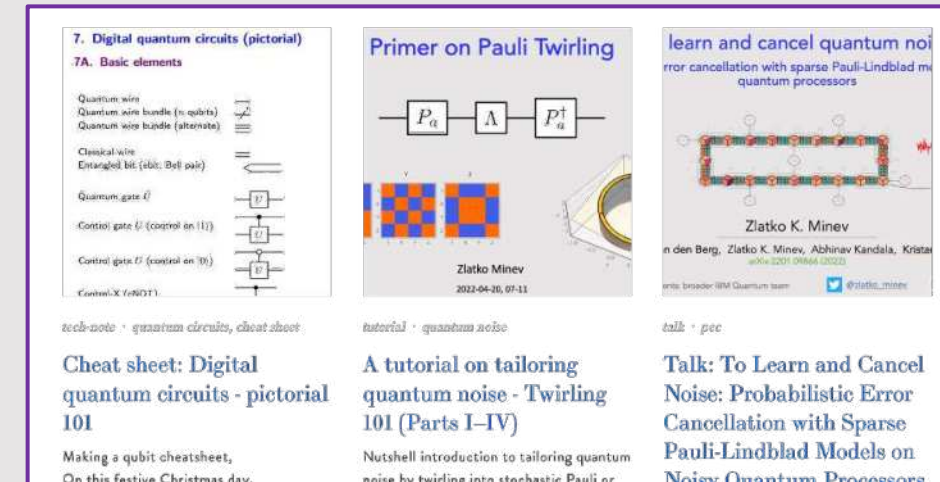
Cheat sheets, Videos, ...

zlatko-minev.com/blog

See also lectures on qiskit.org/learn

Tutorials and additional lecture notes

Latest seminar qiskit.org/events/seminar-series



Have you used
a quantum computer?



Quantum Error Mitigation for Non-Equilibrium Quantum Dynamics

Lecture 1

Big picture

Why quantum computers?

Status and outlook

Why error mitigation?

Noise in quantum computers

Overview of error mitigation

Mitigation fundamentals

Probabilistic error cancelation (PEC)

Introduction

One qubit example

General derivation

Next lectures

Learning noise

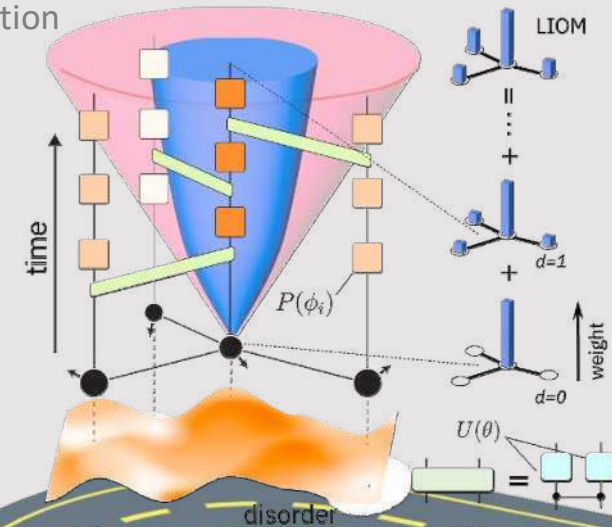
State-of-art PEC experiments

Key techniques: Twirling

T-REX mitigation;

State-of-art experiments at the 120Q+,
depth 50+: uncovering local integrals of
motion

...



Big picture



"Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy."

- R.P. Feynman 1981

International Journal of Theoretical Physics, Vol. 21, Nos. 6/7, 1982

Simulating Physics with Computers

Richard P. Feynman

Department of Physics, California Institute of Technology, Pasadena, California 91107

Received May 7, 1981

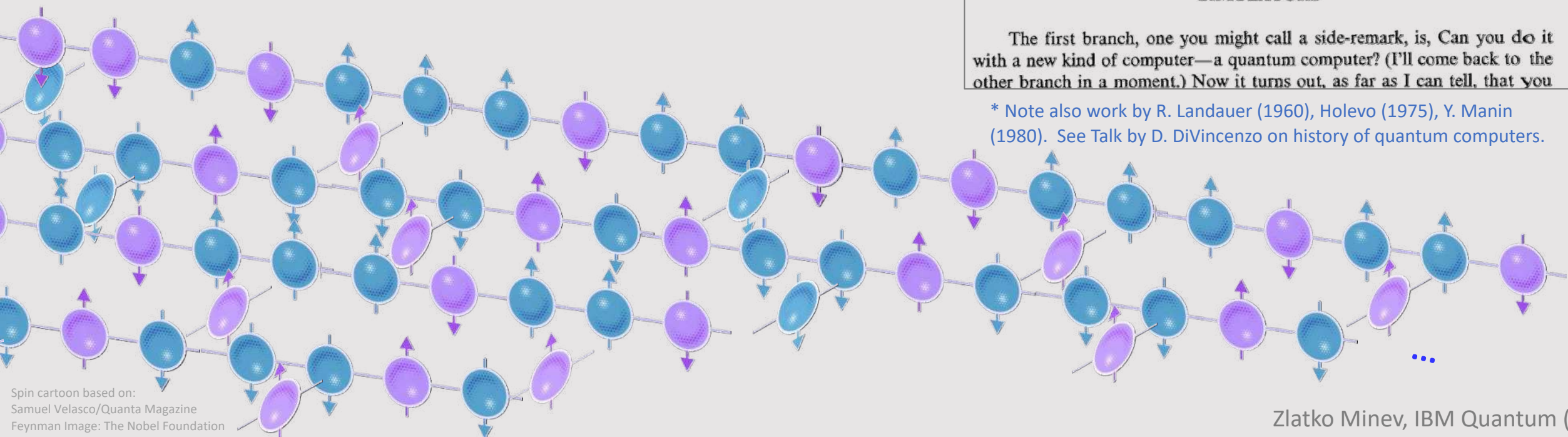
1. INTRODUCTION

On the program it says this is a keynote speech—and I don't know ~~that we thought of before~~ a logical, universal automaton, can we simulate this situation? And I'm going to separate my talk here, for it branches into two parts.

4. QUANTUM COMPUTERS—UNIVERSAL QUANTUM* SIMULATORS

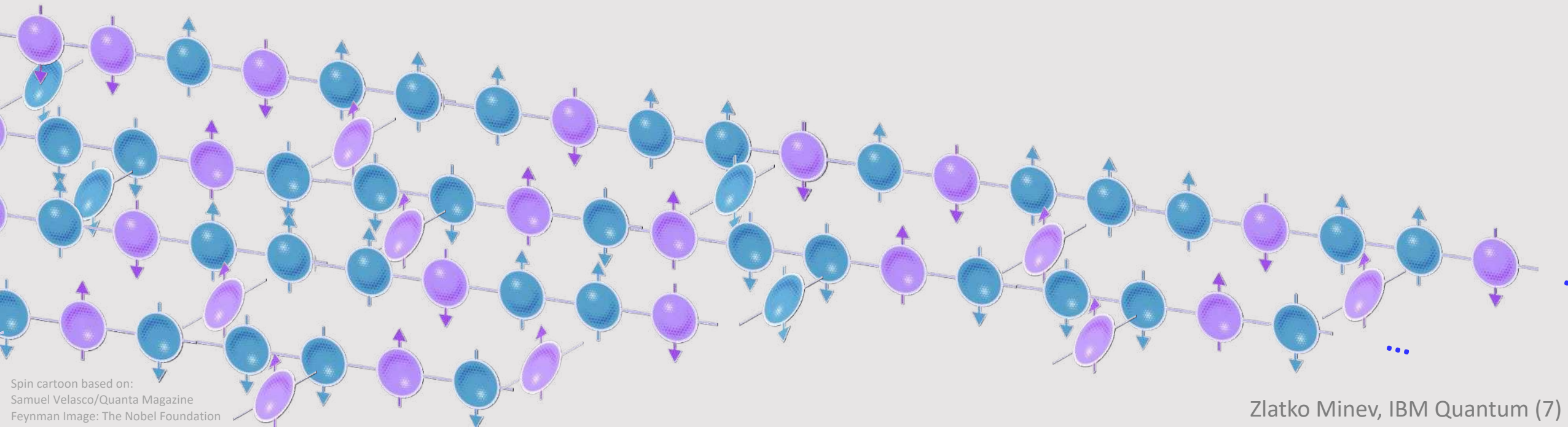
The first branch, one you might call a side-remark, is, Can you do it with a new kind of computer—a quantum computer? (I'll come back to the other branch in a moment.) Now it turns out, as far as I can tell, that you

* Note also work by R. Landauer (1960), Holevo (1975), Y. Manin (1980). See Talk by D. DiVincenzo on history of quantum computers.

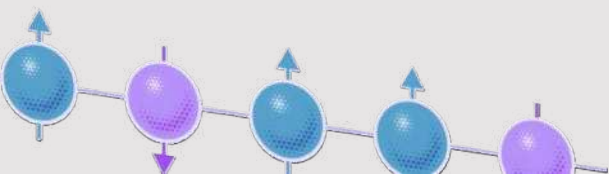


How is it going for quantum computers?

The last 40 years and looking ahead to 10 more
1980 – ... – 2010 – 2020 – 2023 in 60 seconds



I will focus on quantum computers based on **superconducting qubits** (see intro by **Steven Girvin**) since other hardware platforms are covered in the BSS lectures by **Crystal Noel** (next!), **Immanuel Bloch** (this week), **Giulia Semeghini** (earlier), and earlier lectures – and there are a lot of general similarities.



My experience circa 2010

Maybe 1 or 2 qubits
working some small
fraction of the time
in select labs

Photo with dilution fridge called
Sunshine from Michel Devoret's
lab at Yale during my Ph.D.



Hopes for a
working
qubit in
here

This year 2023

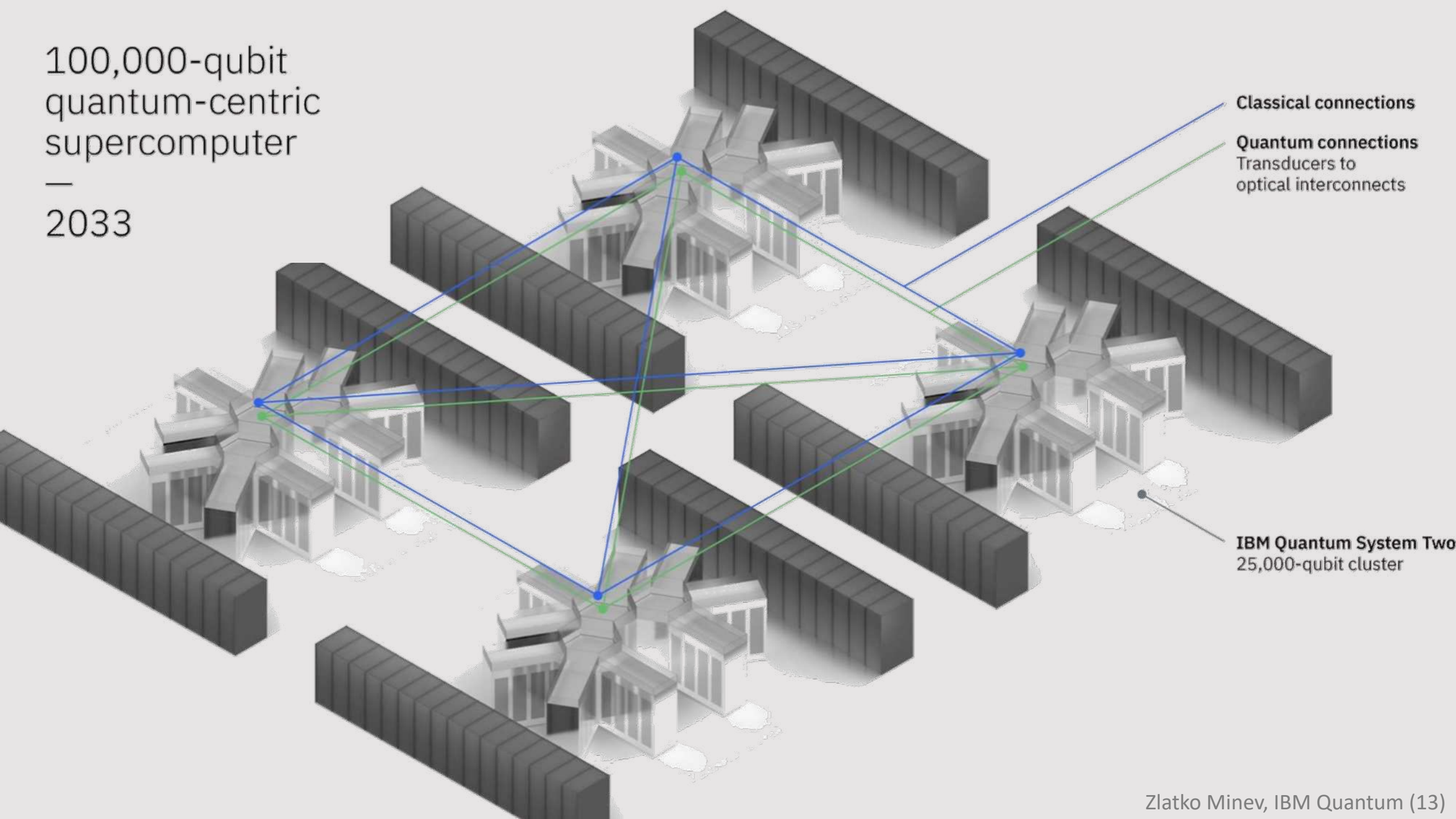
A 127-qubit quantum computer installed in the the lobby cafeteria of a research building dutifully executing jobs almost all the time.







100,000-qubit
quantum-centric
supercomputer
—
2033

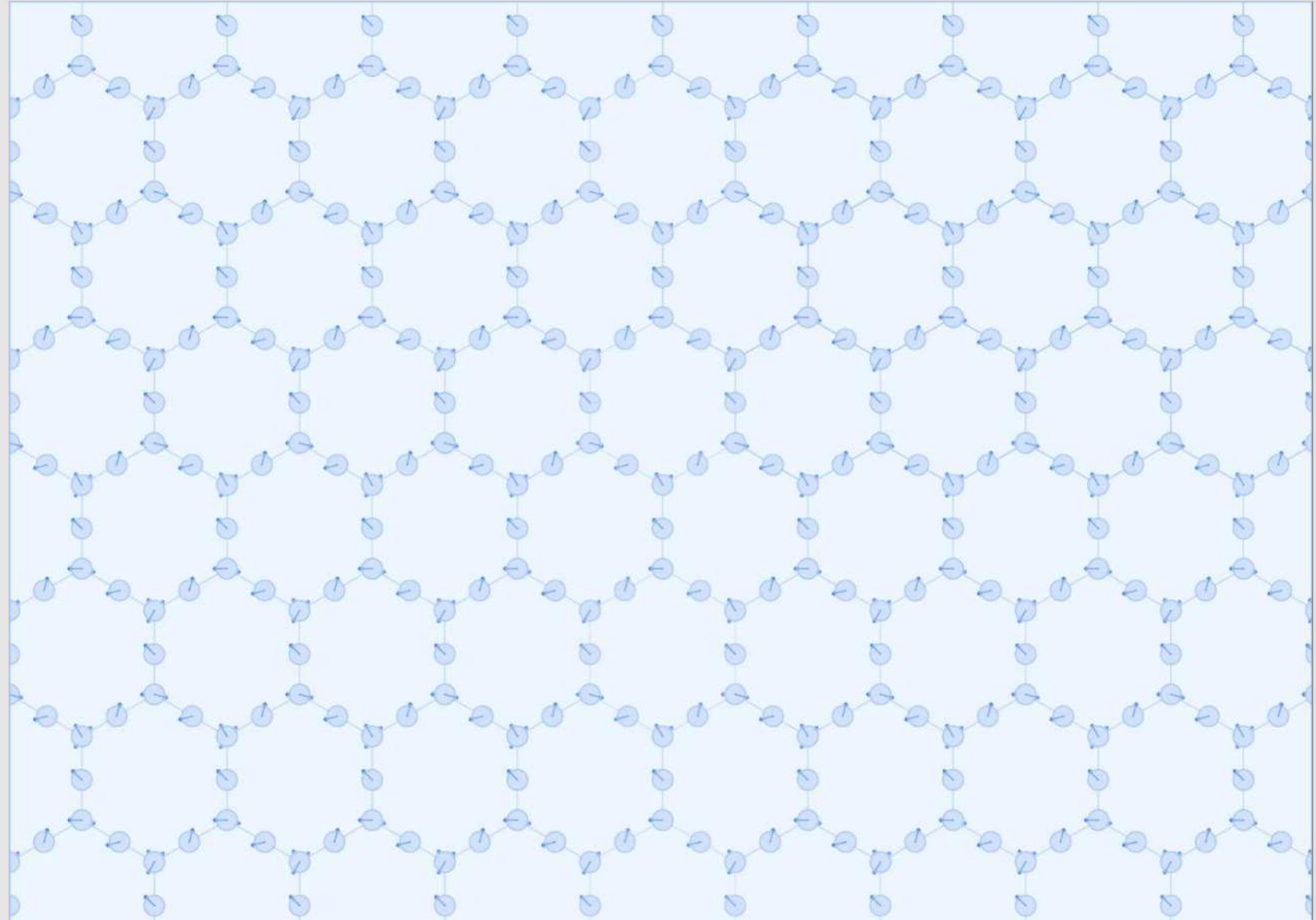


Back to today ...

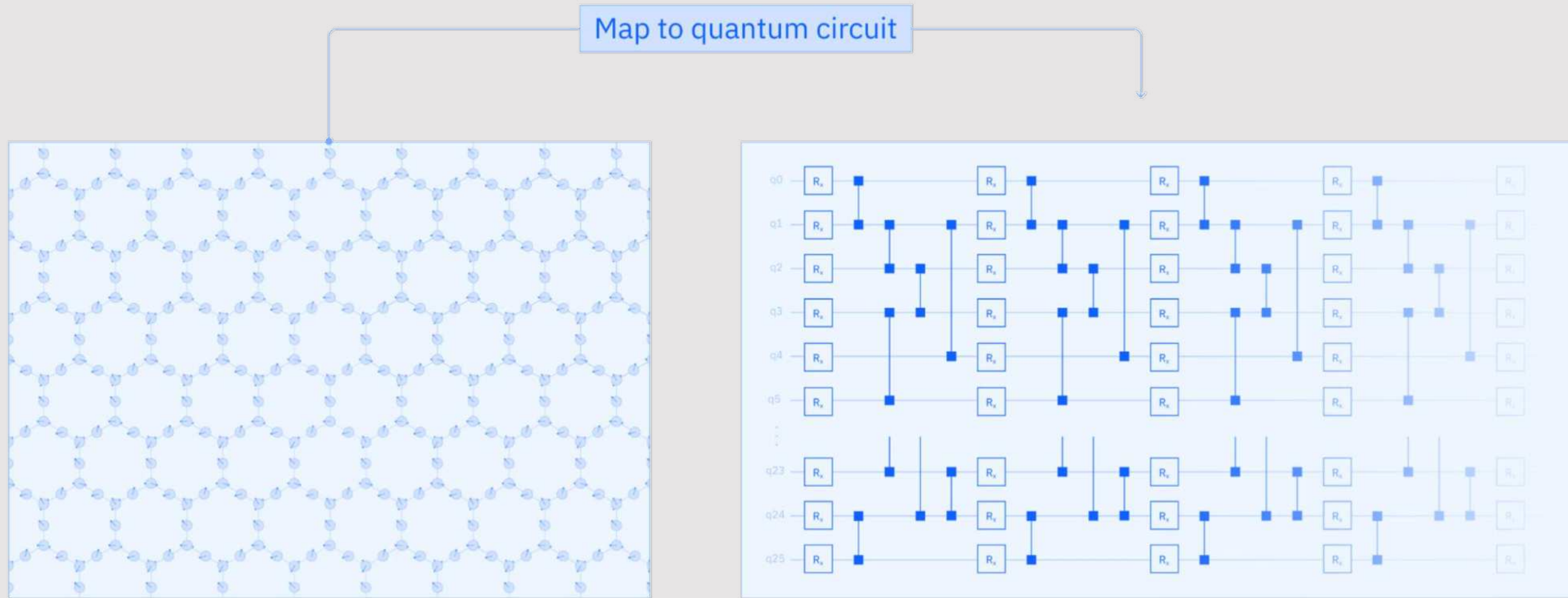
Non-equilibrium quantum simulation

Example task:

Simulate the out-of-equilibrium quantum dynamics of a 2D spin chain lattice to find the evolution of the global and local magnetization.

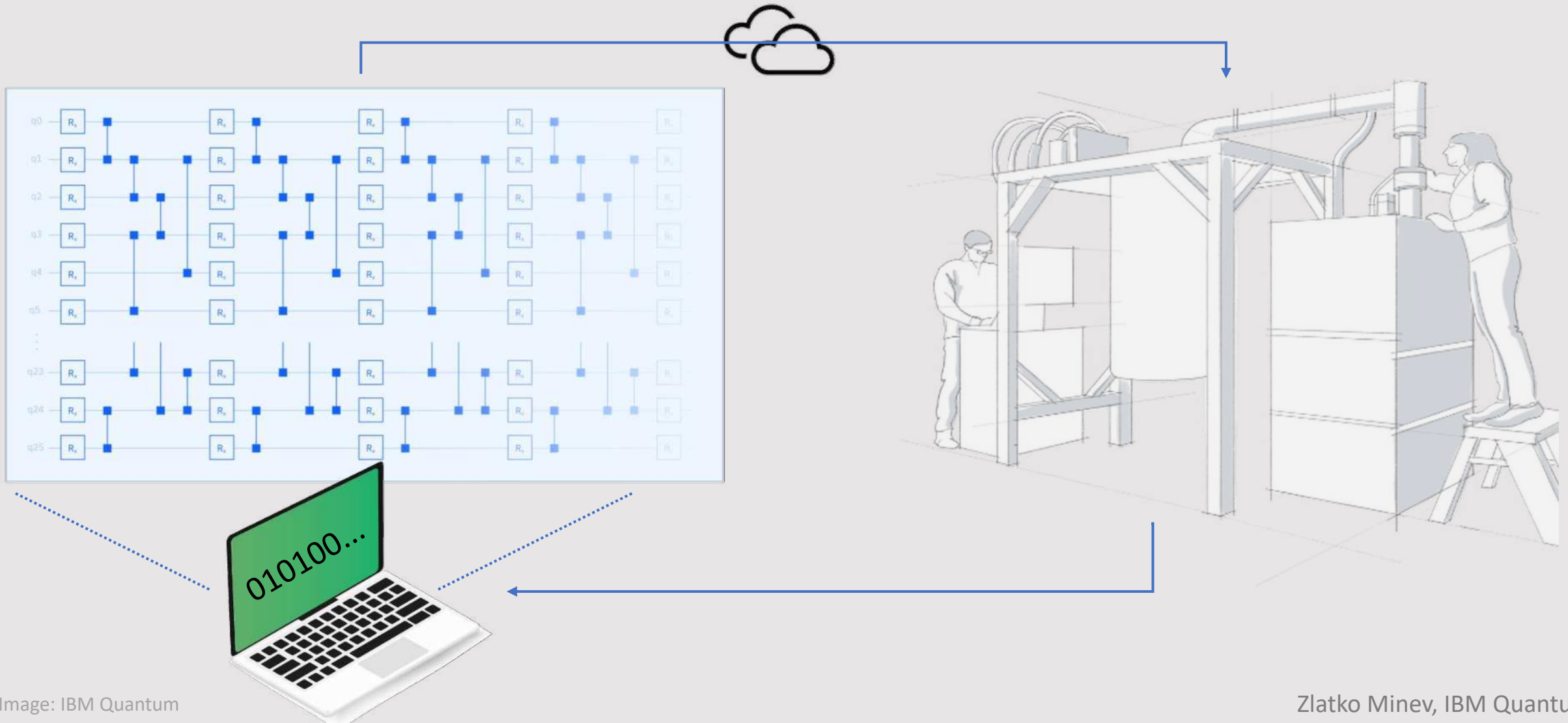


Quantum simulation on a quantum computer



Quantum simulation on a quantum computer

Execute on a real quantum computer device and obtain results as classical data



Biggest challenge?

Please do share



Biggest challenge

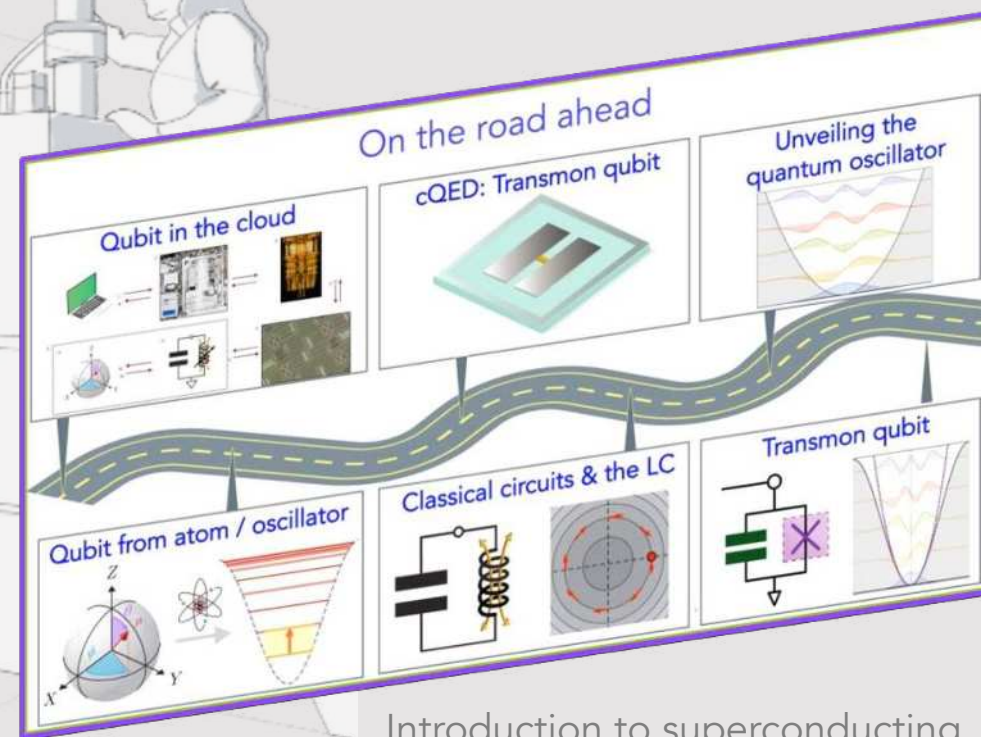
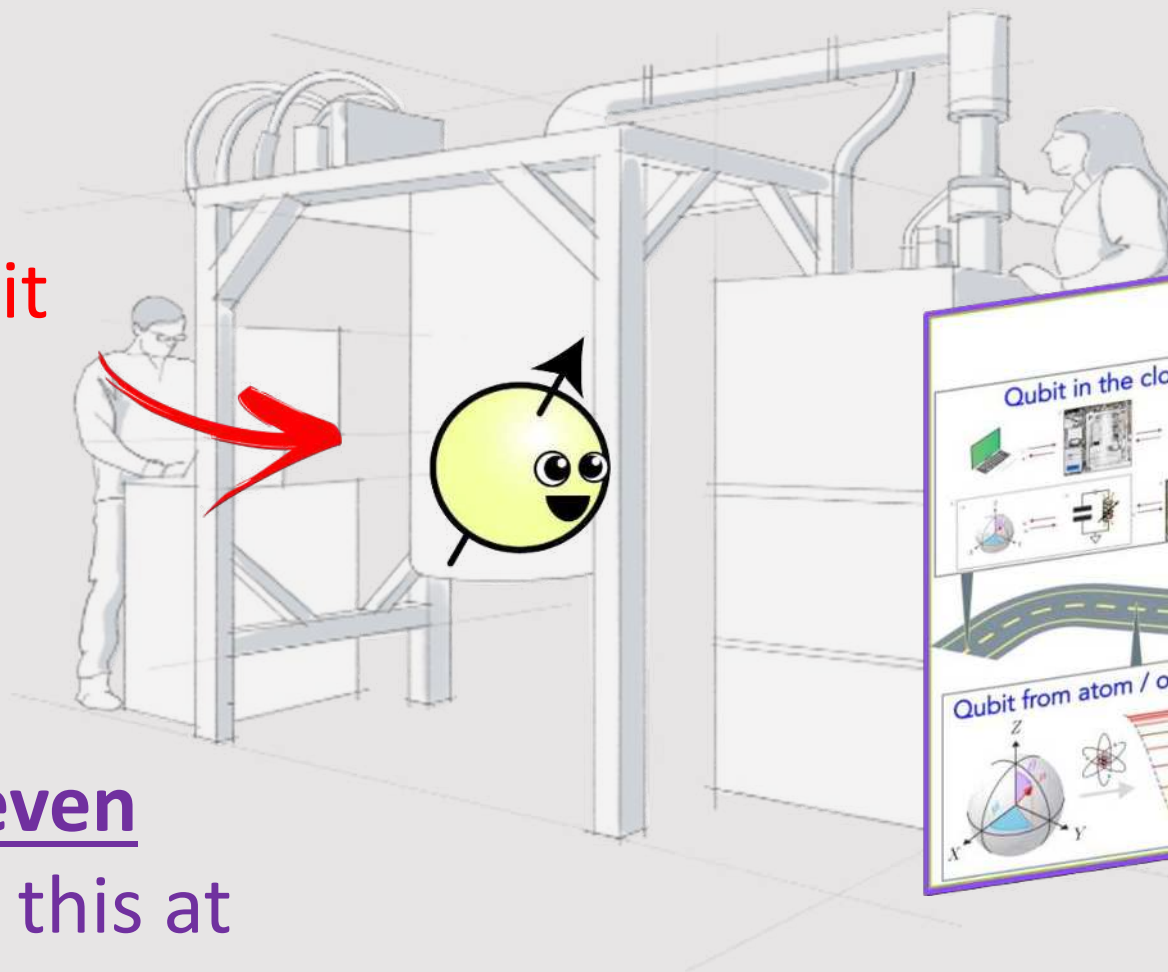
Noise
(Errors)



Hello World with a real experiment!



A qubit



Introduction to superconducting qubits (cQED)

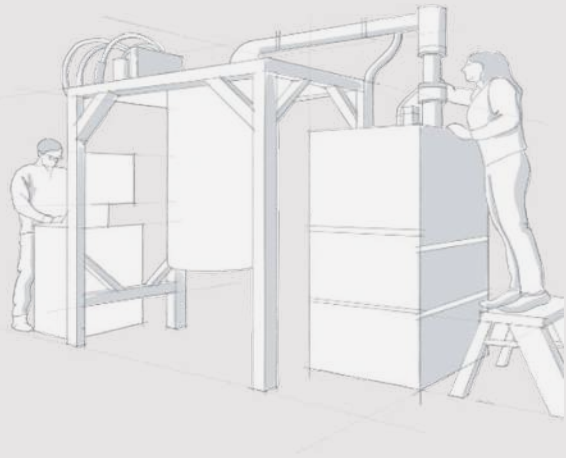
Lecs. 16-21 Minev

QGSS 2020 at qiskit.org/learn

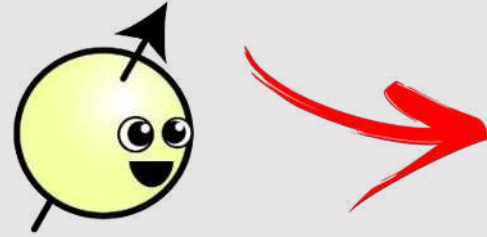
Zlatko Minev, IBM Quantum (21)

See lectures by Steven Girvin right before this at BSS23 for cQED!

Hello World! building blocks

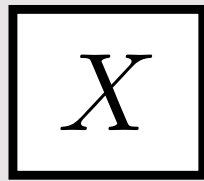


A qubit

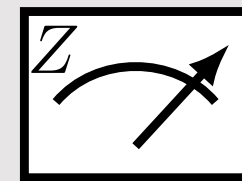
 $|1\rangle$ $|0\rangle$

Computational
basis states

Operations: qubit gate



Measurements: qubit observable



refresher:

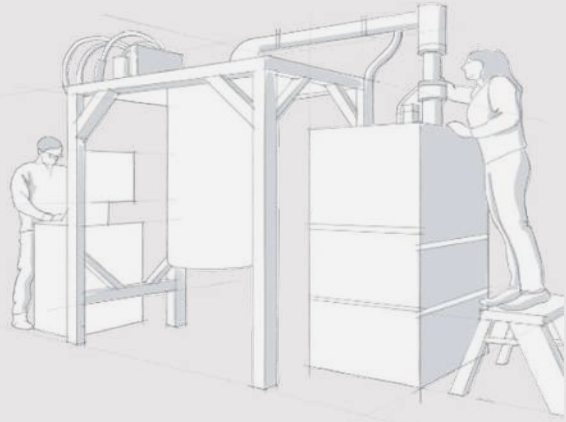
$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

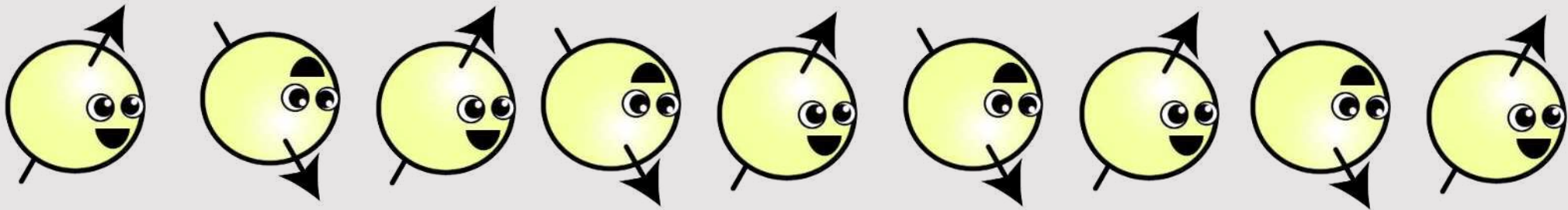
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! Even-odd algo: qubit flipper



Task: Classify or report if a classical positive integer d is even or odd.



flip spin d times, measure polarization

refresher:

$$X |0\rangle = |1\rangle$$

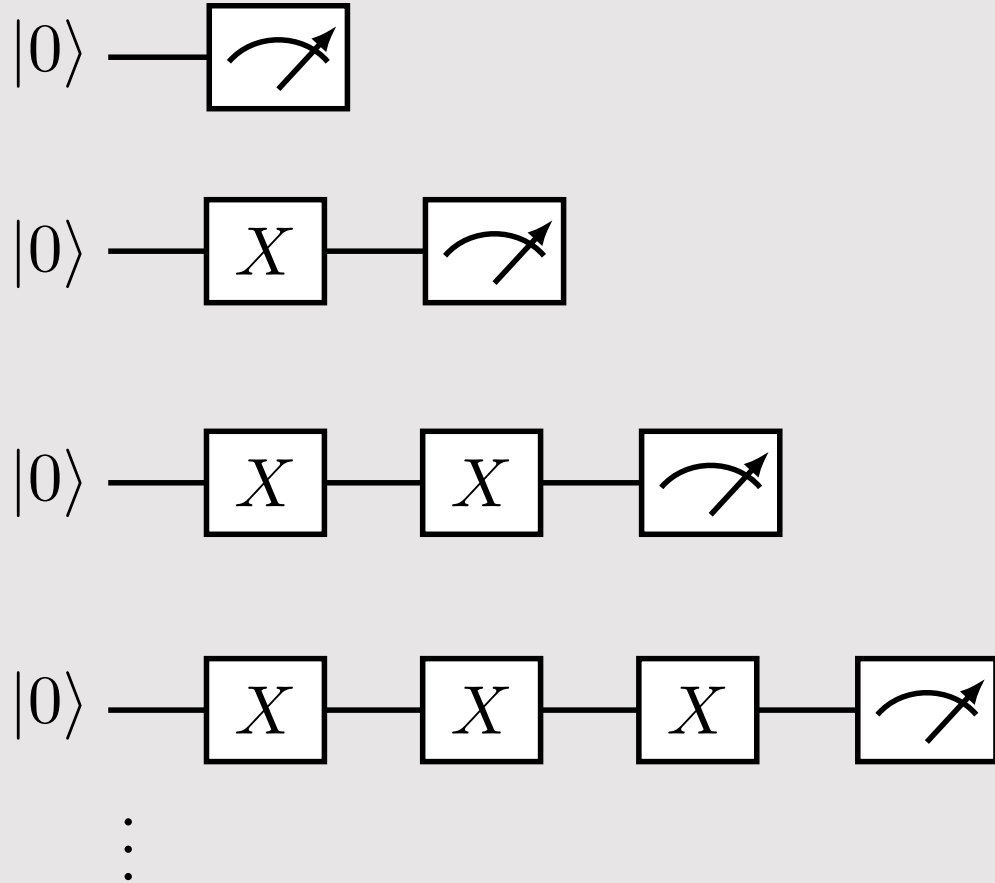
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! qubit flipper quantum circuits

depth



refresher:

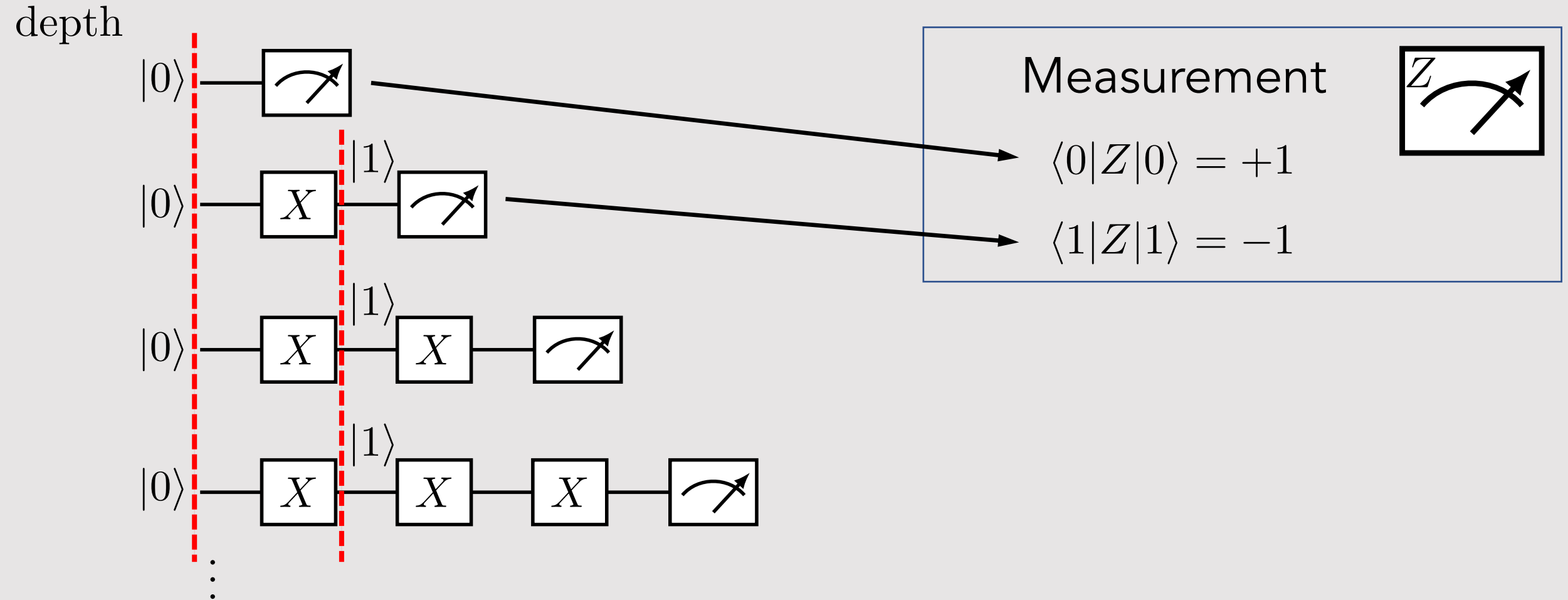
$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! "debugger" step through



refresher:

$$X|0\rangle = |1\rangle$$

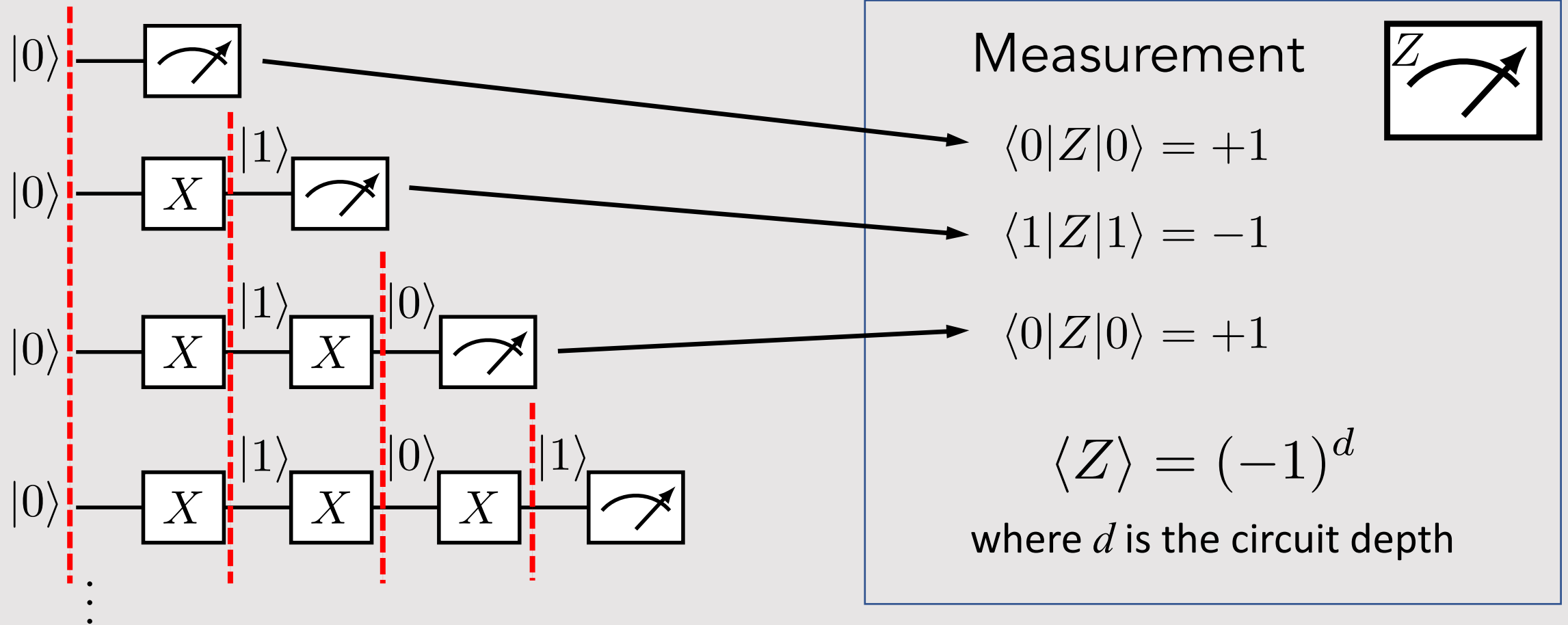
$$X |1\rangle = |0\rangle$$

$$Z \left| 0 \right\rangle = +1 \left| 0 \right\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! “debugger” step through

depth



refresher:

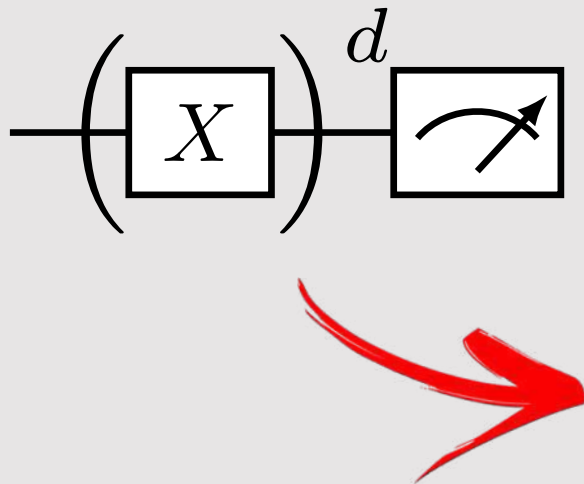
$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

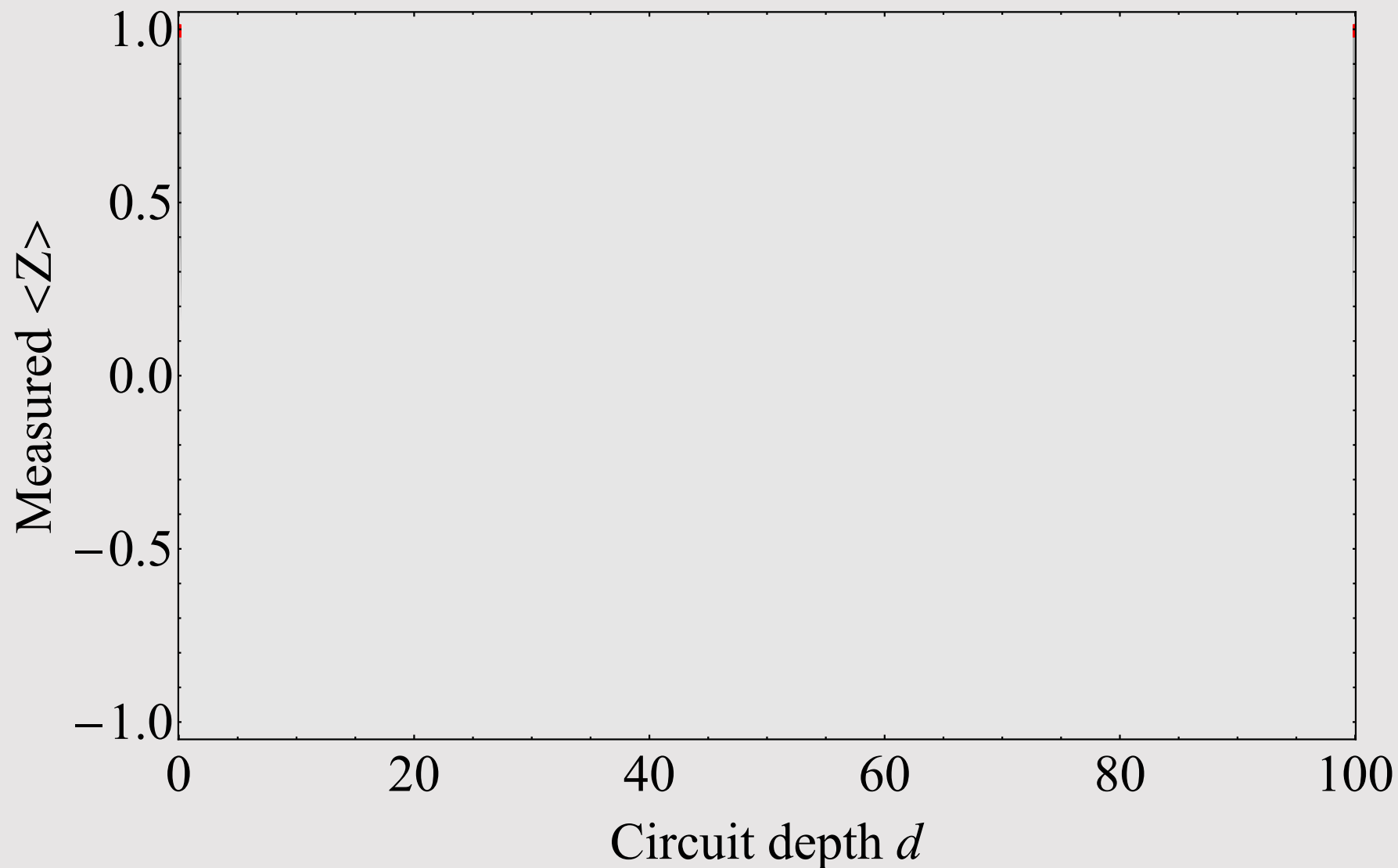
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! Ideal expectation results

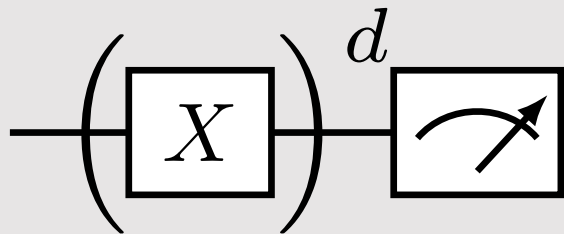


$$\langle Z \rangle = (-1)^d$$

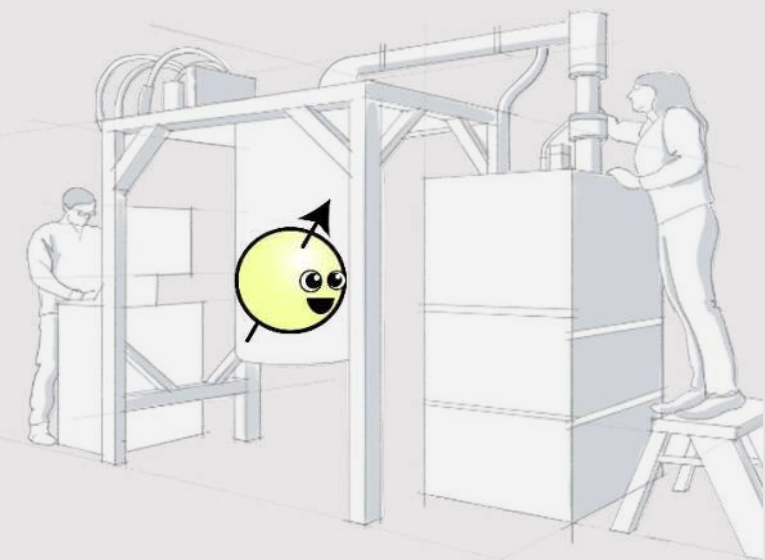


Hello World! Ideal expectation results

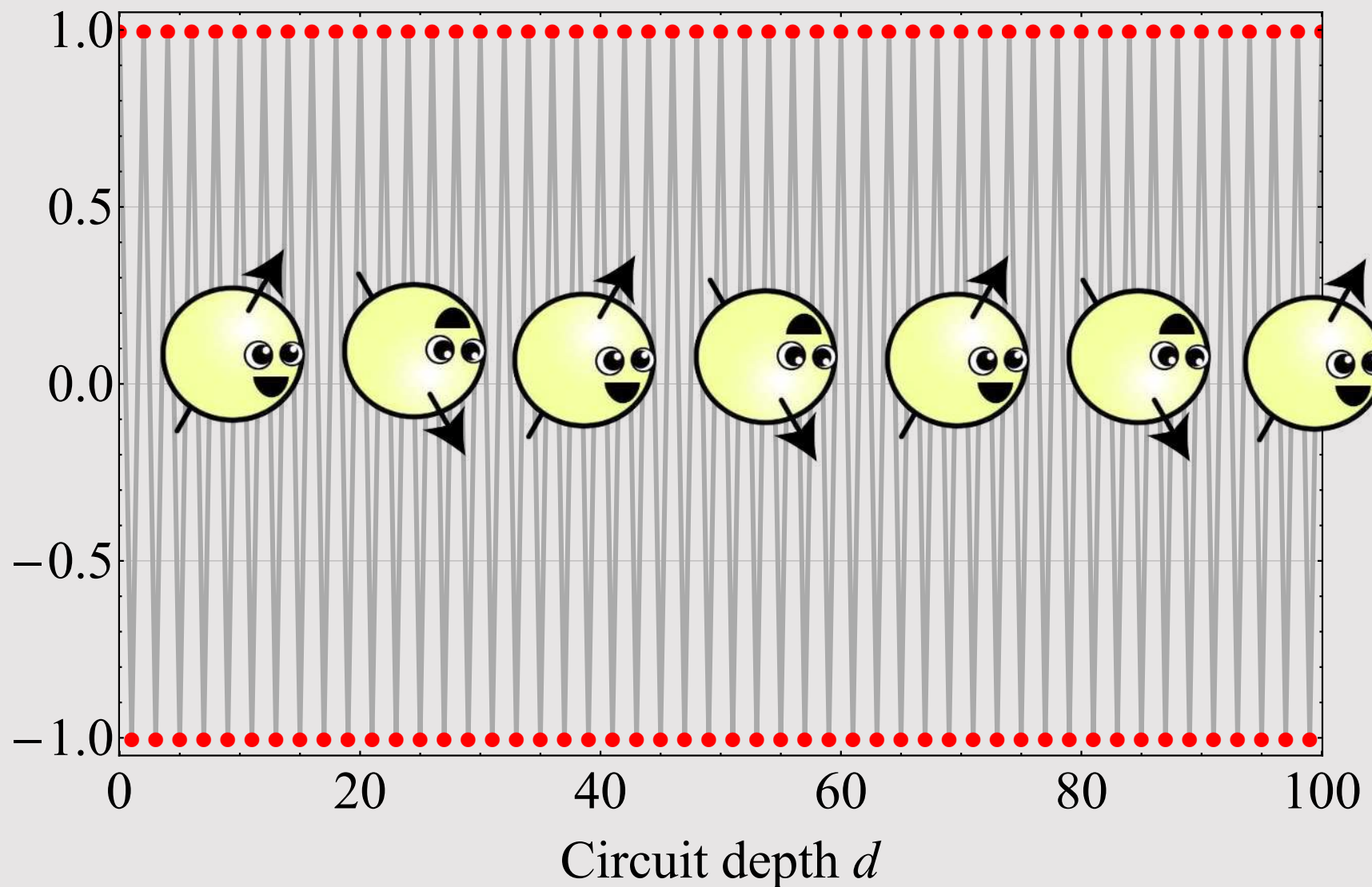
$$\langle Z \rangle = (-1)^d$$



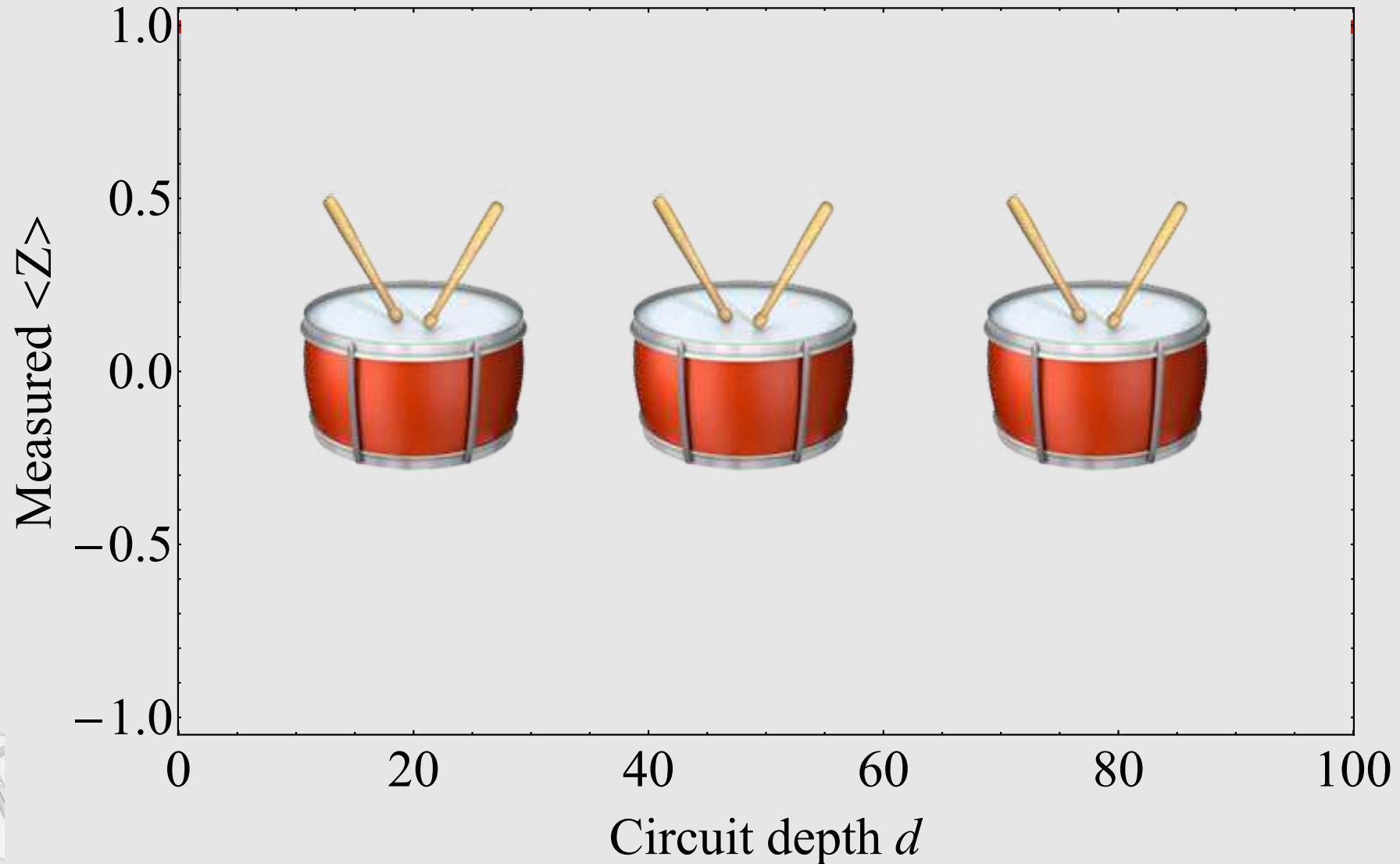
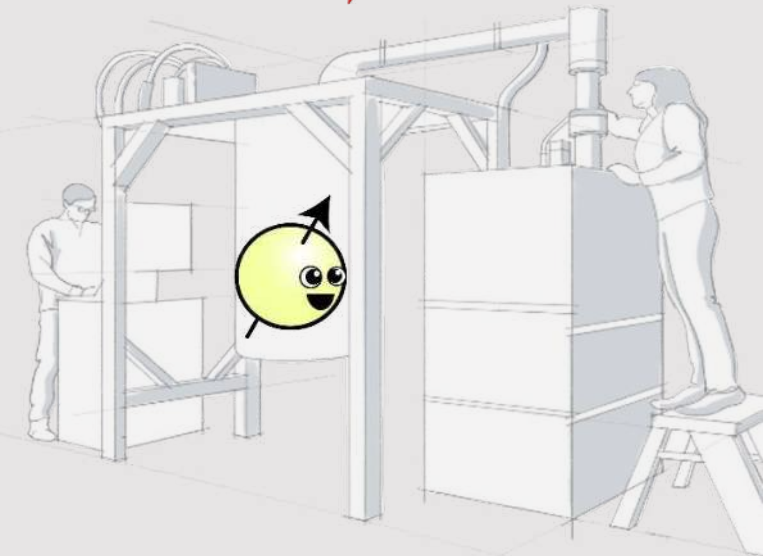
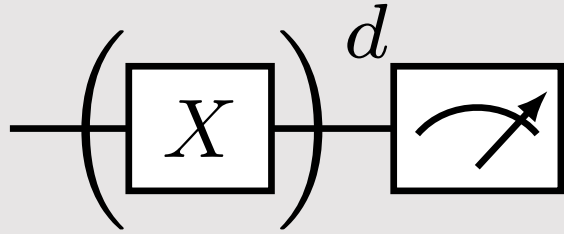
Let's run on a real device!



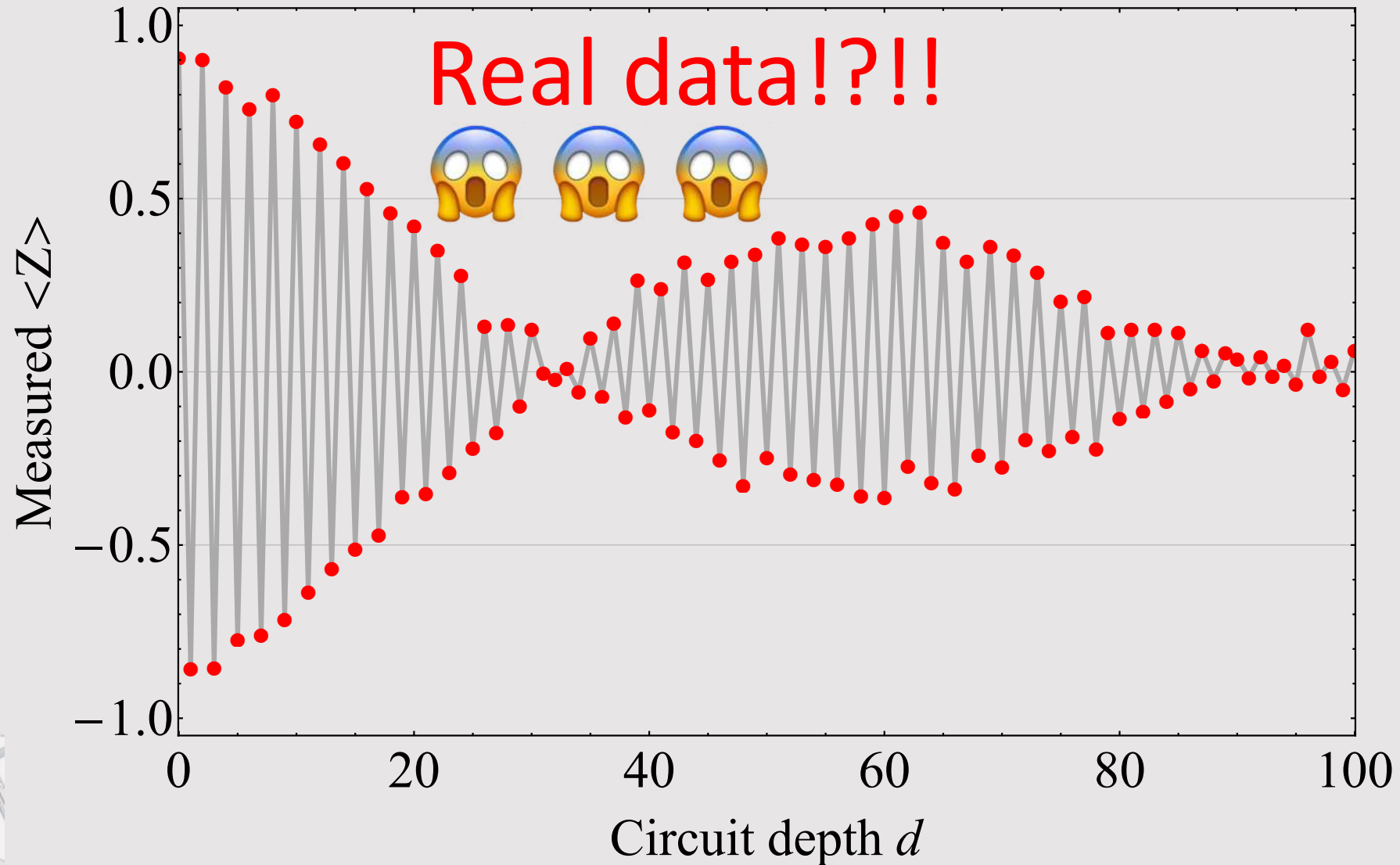
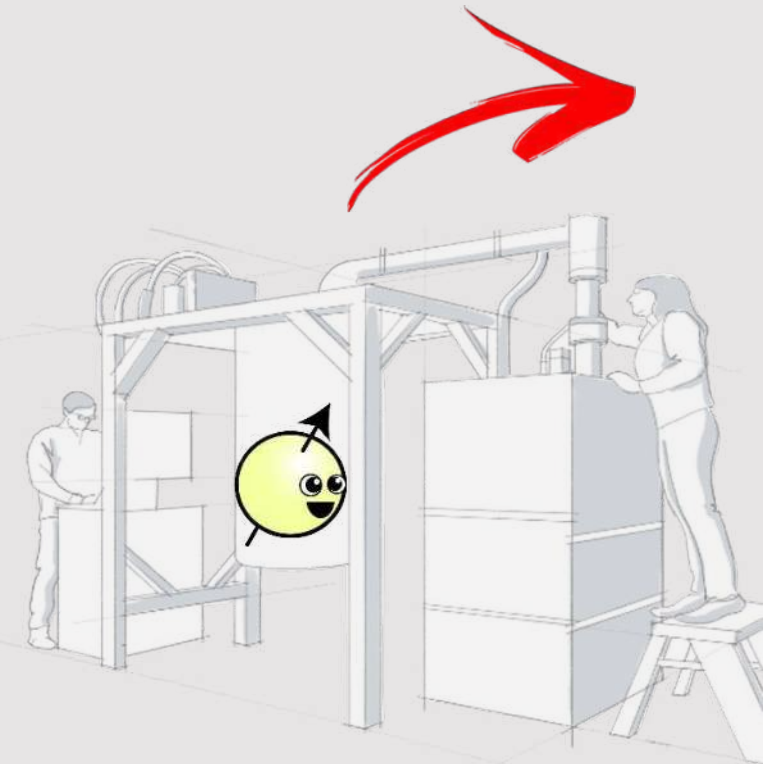
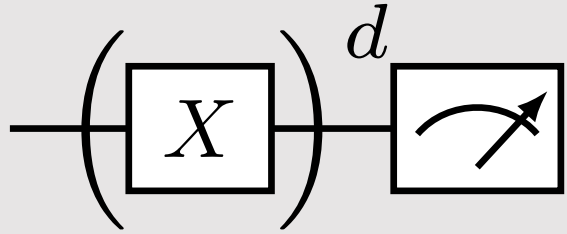
Measured $\langle Z \rangle$



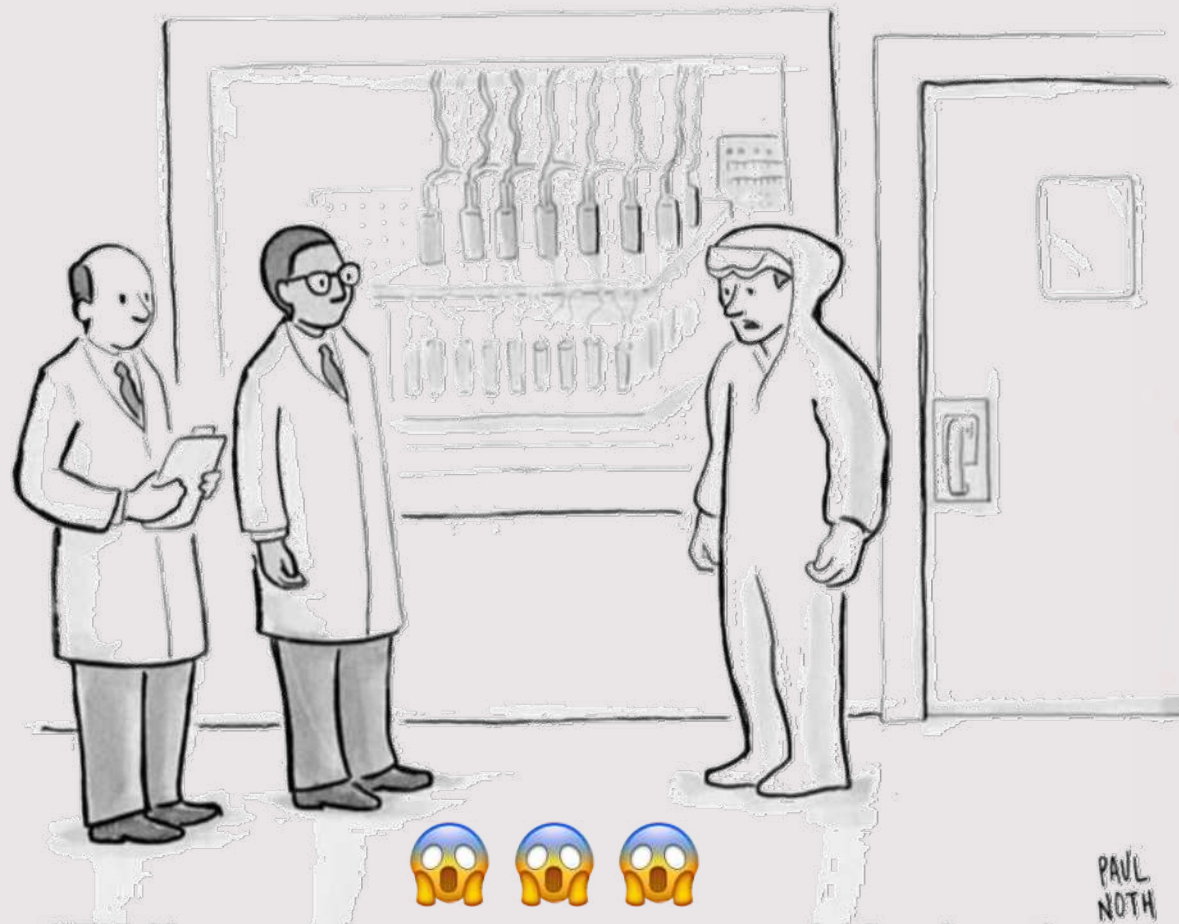
Hello World! Running



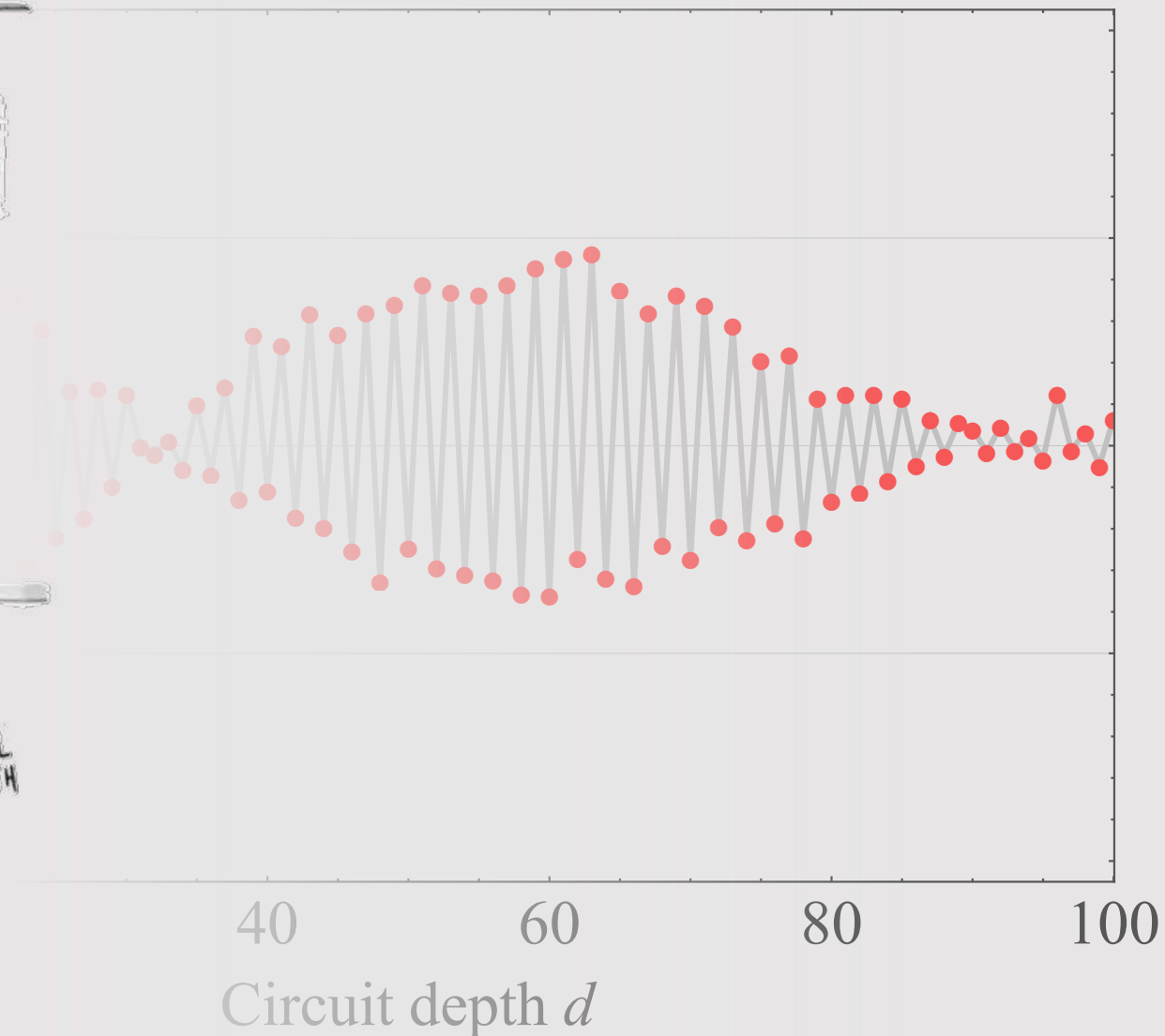
Hello World! Real expectation results

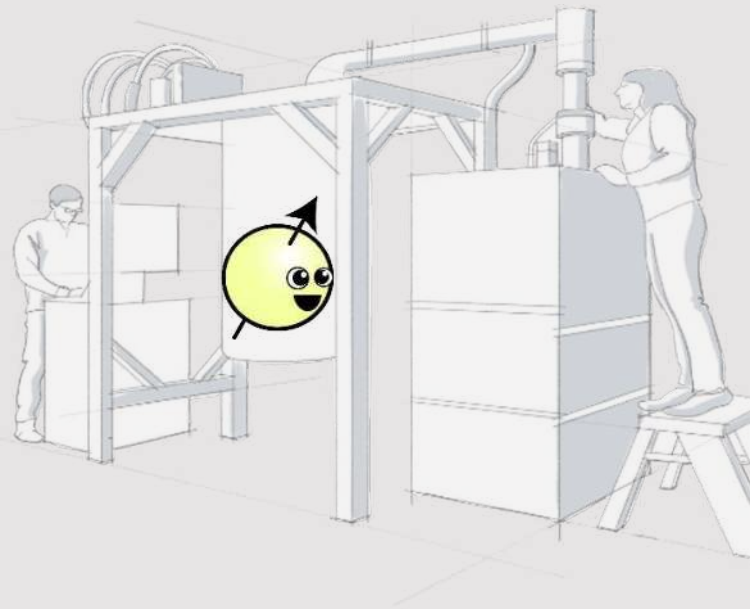


Real & noisy quantum processors: Why study noise?



“Well, your quantum computer is broken in every way possible simultaneously.”

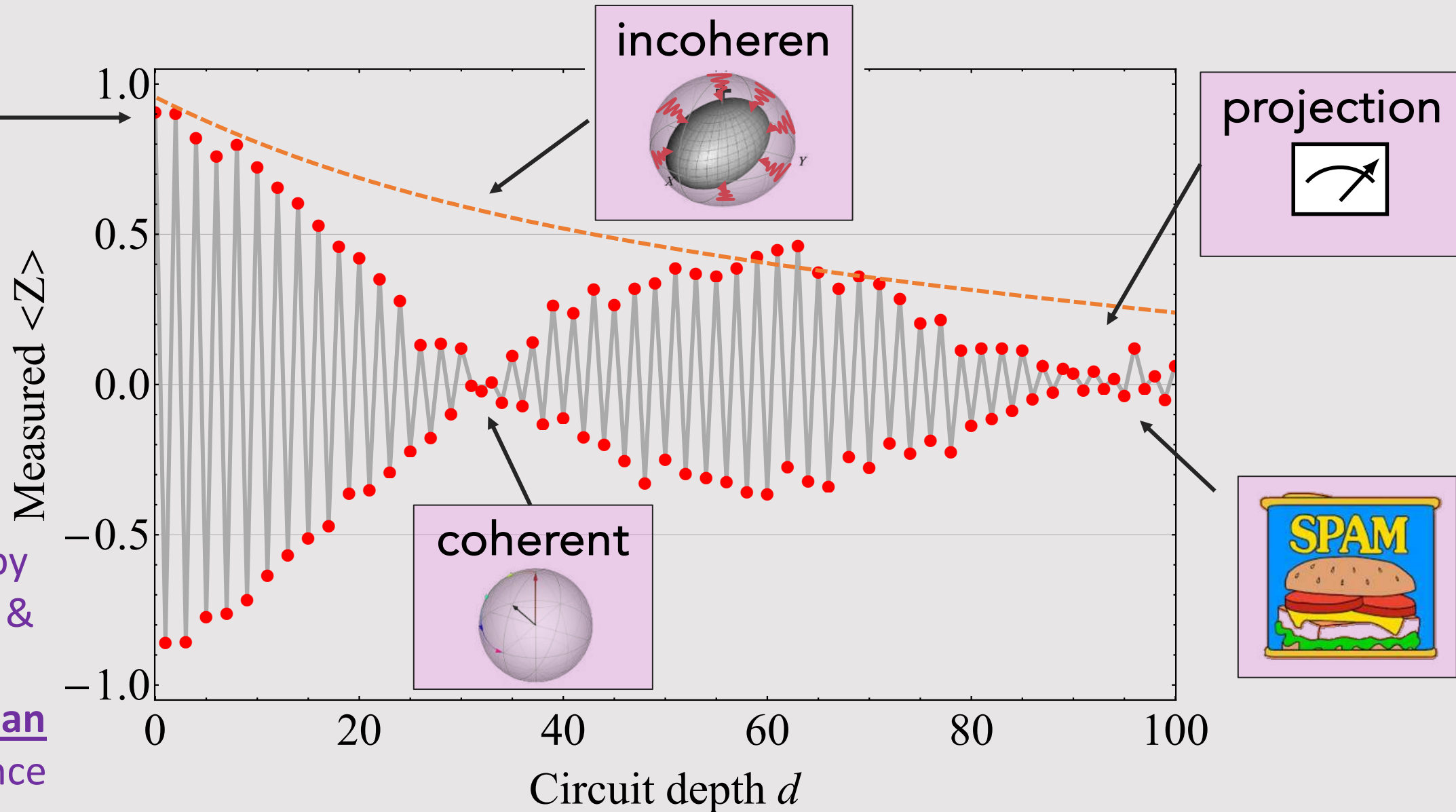




"Quantum phenomena
do *not* occur in a Hilbert space,
they occur in a laboratory."

Asher Peres

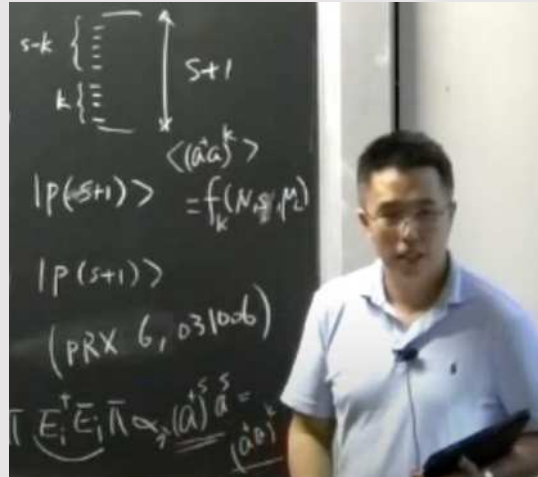
Elements of 🤯 noise



See lectures by
[Ehud Altman](#) &
[Sarang Gopalakrishnan](#)
on decoherence
at BSS23

How to deal with errors due to noise?

Monitor
Error occurs
Error detect



See lectures by Liang Jiang, Victor Albert, and Aleksander Kubica at BSS23 for QEC!

Quantum error correction

Shor, PRA (1995), ...

How to deal with errors due to noise?

Monitor
Error occurs
Error detect



Quantum error correction

Shor, PRA (1995), ...

Monitor
Error anticipated
Tell signal detected



Catch and reverse

Minev, Nature (2019), ...

No monitor
Error occurs
Error undetected

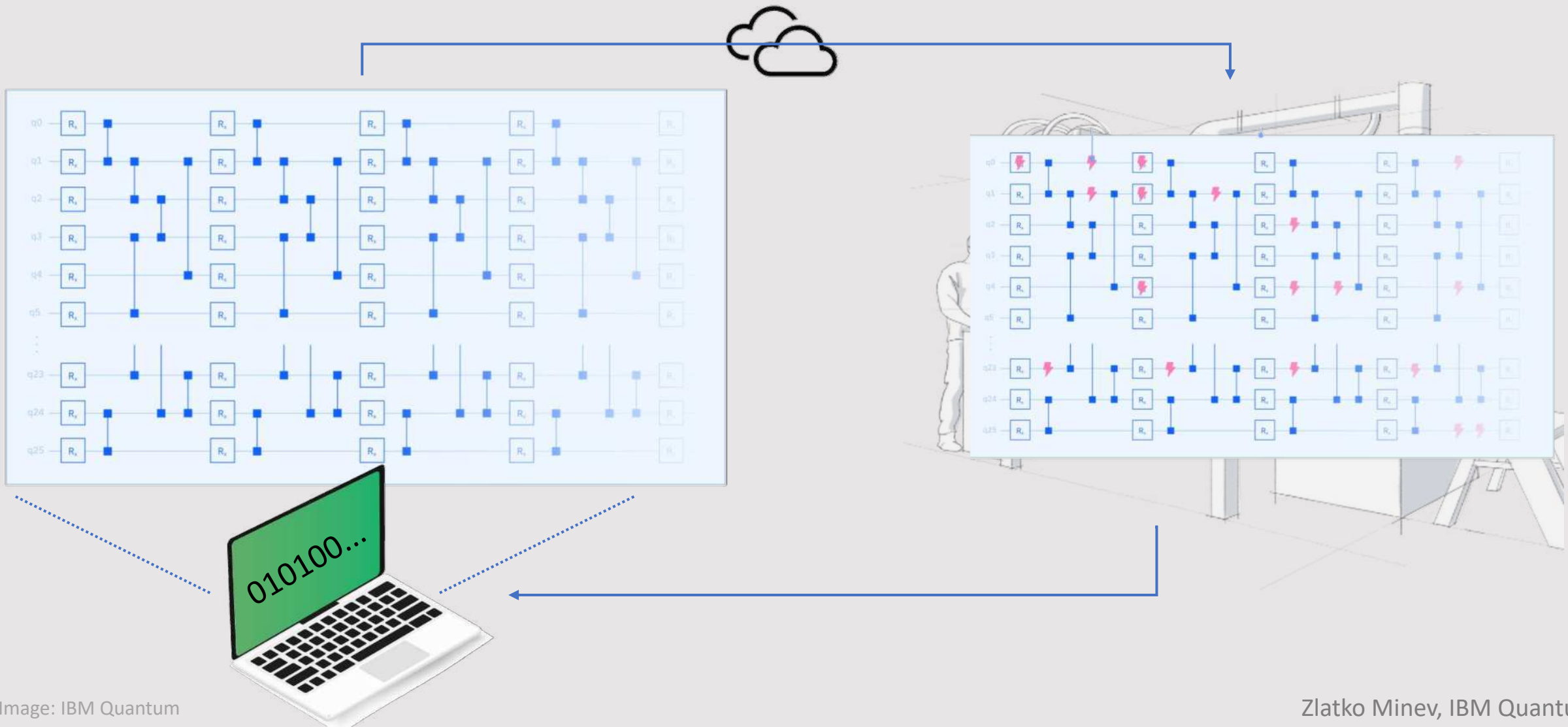


Error mitigation

... subject of today

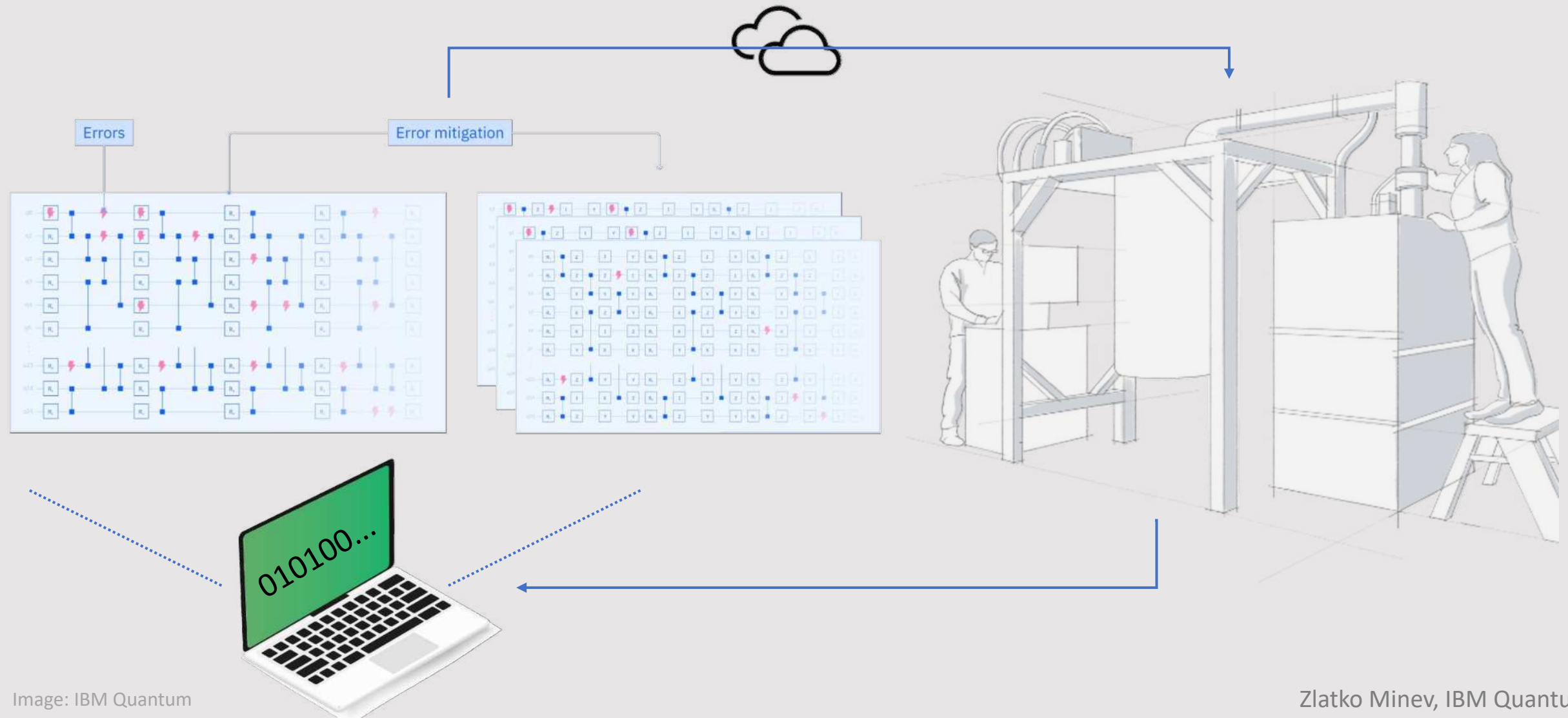
Quantum simulation on a noisy quantum computer

Execute on a real quantum computer device and obtain results as classical data

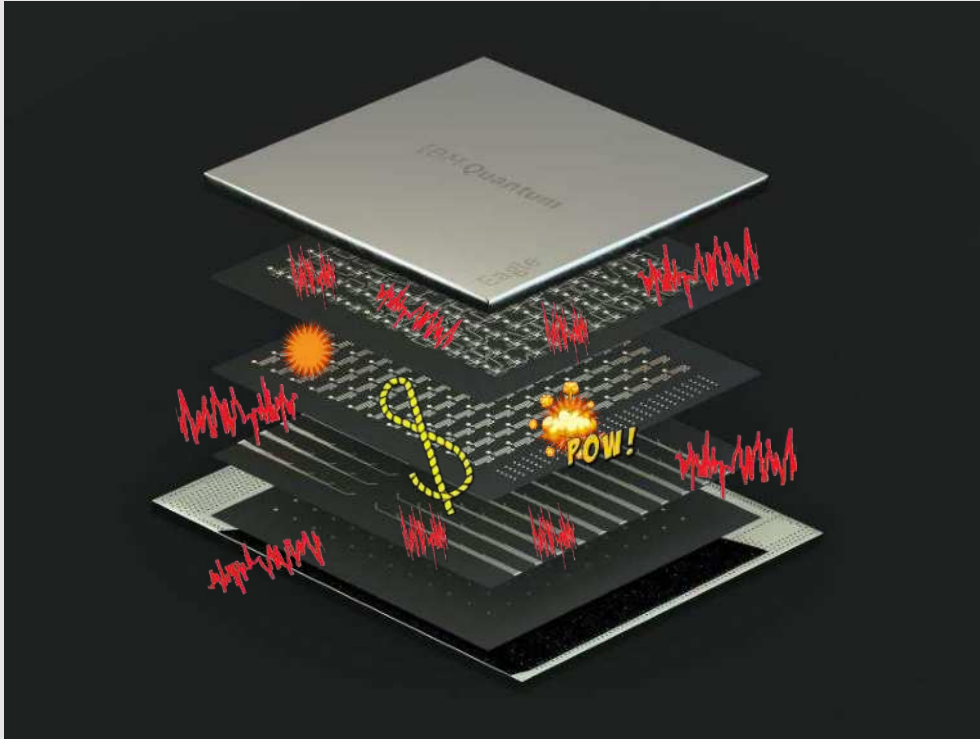


Quantum error mitigation overview

Execute on a real quantum computer device and obtain results as classical data



Error mitigation and error correction



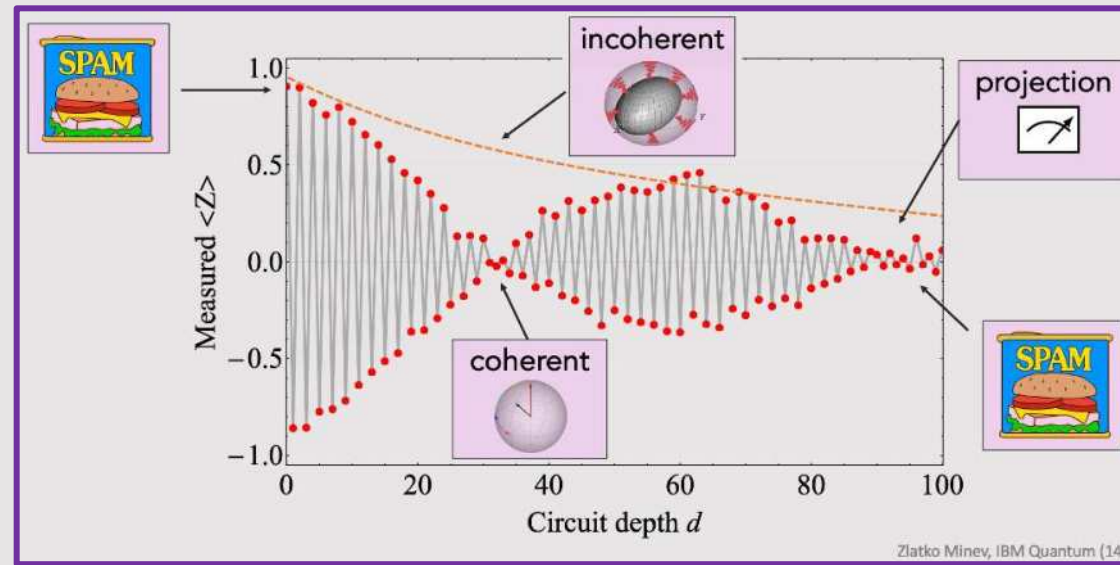
Error mitigation: working with what you have

- **benefit** suppress errors on classical results (expectation values)
- **q-cost** no extra qubits or hardware resources needed
- **c-cost** trades classical resources (post-processing) for lower error
- **limitation** bad asymptotic scaling: high number of samples & circuits

Error correction: protecting quantum information

- **benefit** suppress & correct errors to arbitrarily small level
- **q-cost** very large qubit and hardware overhead
- **c-cost** decoding and encoding can be classically costly
- **challenge** requires fault-tolerant operations and readout

Error mitigation landscape



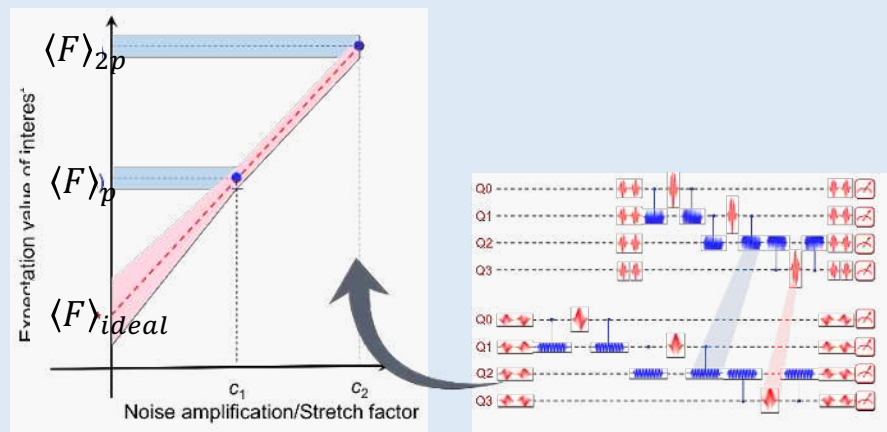
more
speed

more information,
accuracy



Error mitigation landscape

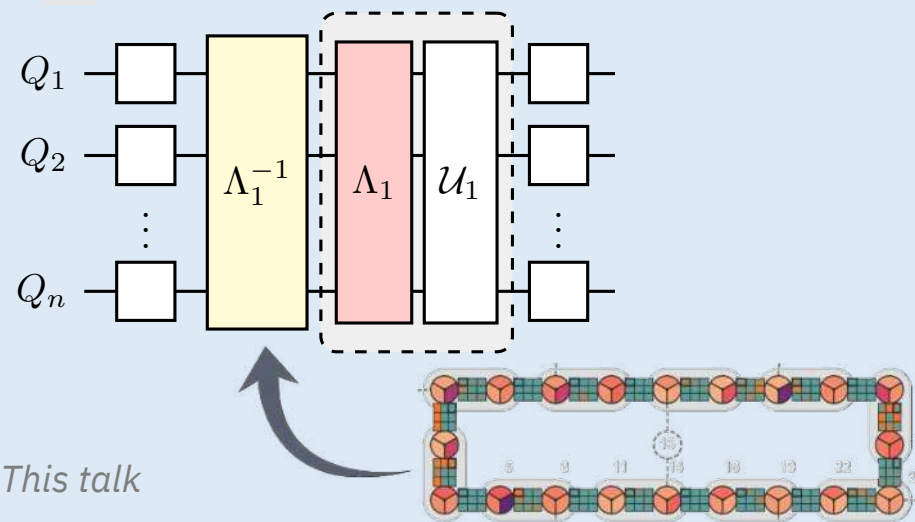
Zero-noise extrapolation (ZNE)



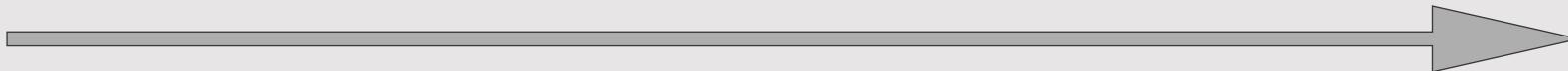
Nature 567, 491 (2019)

40

Probabilistic error cancellation (PEC)



more
speed



more information,
accuracy

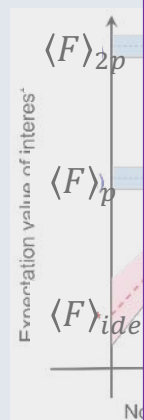


Zlatko Minev, IBM Quantum (40)

Error mitigation landscape

Zero-noise extrapolation (ZNE)

Probabilistic error cancelation (PEC)



Qiskit qiskit.org Overview Learn Community

English

Search Qiskit Runtime IBM

Get started

- Overview
- Getting Started
- backend.run vs. Qiskit Runtime
- Introduction to primitives

Tutorials

- Get started with Estimator
- Get started with error suppression and error mitigation
- VQE with Estimator
- CHSH with Estimator
- Get started with Sampler

Qiskit Runtime IBM Client documentation > Error suppression and error mitigation with Qiskit Runtime

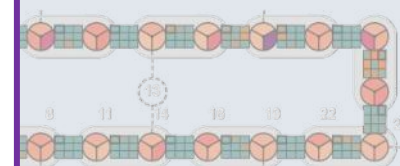
NOTE

This page was generated from [docs/tutorials/Error-Suppression-and-Error-Mitigation.ipynb](#).

Error suppression and error mitigation with Qiskit Runtime

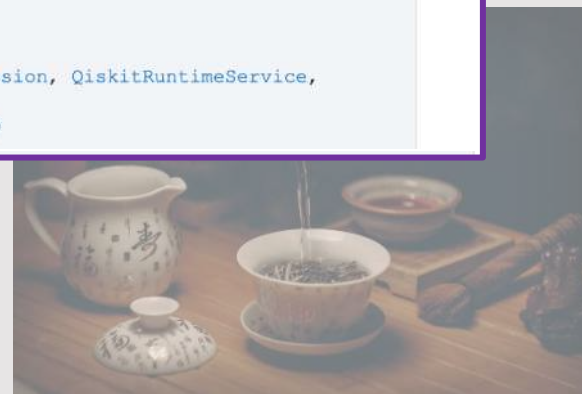
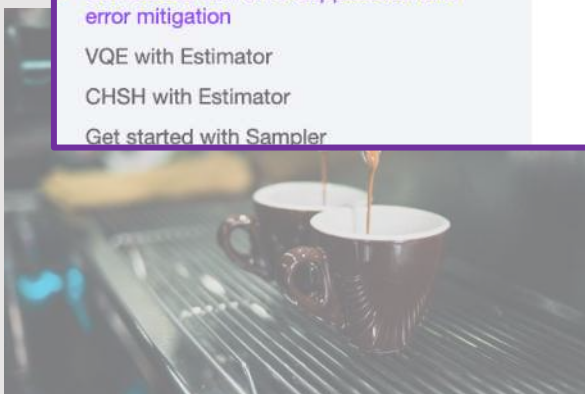
```
[1]: import datetime
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
plt.rcParams.update({"text.usetex": True})
plt.rcParams["figure.figsize"] = (6,4)
mpl.rcParams["figure.dpi"] = 200

from qiskit_ibm_runtime import Estimator, Session, QiskitRuntimeService,
Options
from qiskit.quantum_info import SparsePauliOp
from qiskit import QuantumCircuit
```



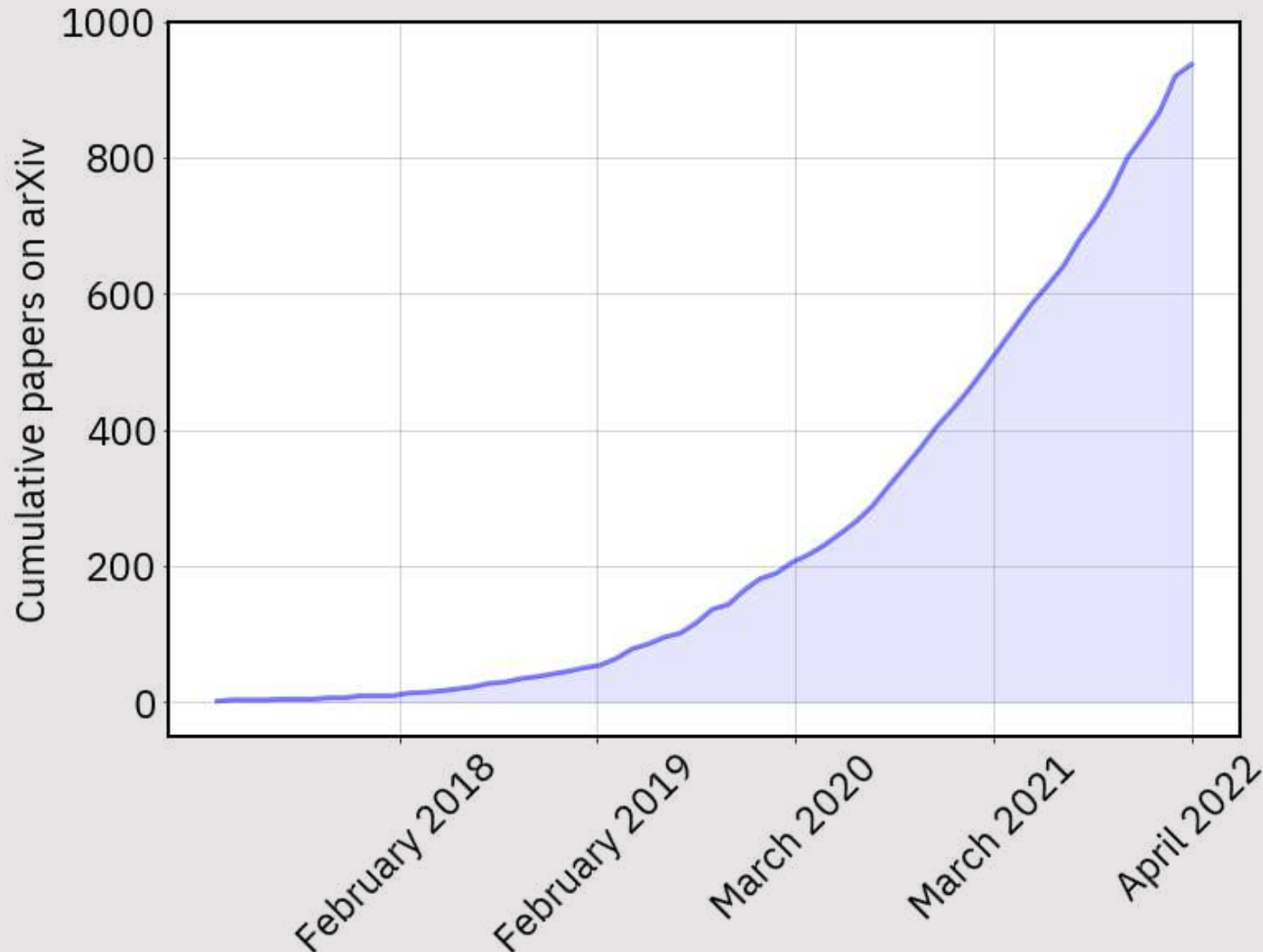
more
speed

more information,
accuracy



Adoption of error mitigation

Papers involving error mitigation over time



Examples

ARTICLE <https://doi.org/10.1038/s41467-020-14376-z> OPEN

Error-mitigated quantum gates exceeding physical fidelities in a trapped-ion system

Shuaining Zhang¹, Yao Lu¹, Kuan Zhang^{1,2}, Wentao Chen¹, Ying Li^{3*}, Jing-Ning Zhang^{1,4*} & Kihwan Kim^{1*}

Matrix product channel: Variation to mitigate noise and reduce errors

Sergey Filippov,^{*} Boris Sokolov, Ma Borrelli, Daniel Cavalcanti, Sabrina Algorithmiq Ltd, Kanavakat

Article | Published: 08 May 2023

Probabilistic error cancellation with sparse Pauli-Lindblad models on noisy quantum processors

Ewout van den Berg, Zlatko K. Minev, Abhinav Kandala & Kristan Temme

Nature Physics (2023) | Cite this article

Quantum Error Mitigation

Zhenyu Cai,^{1,2,*} Ryan Babbush,³ McClean,³ and Thomas E. O'Brien,¹ Department of Materials, University of California, Santa Barbara, CA 94949, USA
²Quantum Motion, 9 Sterling Way, San Francisco, CA 94133, USA
³Google Quantum AI, Venice, California 90293, USA
⁴NTT Computer and Data Science Graduate School of China Academy
(Dated: July 3, 2023)

npi | quantum information

ARTICLE OPEN

Fundamental limits of quantum error mitigation

Ryuji Takagi^{1,2}, Suguru Endo^{2,3}, Shintaro Minagawa^{3,4} and Mile Gu^{1,4,5}

PHYSICAL REVIEW LETTERS **127**, 200505 (2021)

Error Mitigation for Universal Gates on Encoded Qubits

Christophe Piveteau
IBM Quantum, IBM Research—Zürich, 8803 Rüschlikon, Switzerland

Single-shot error mitigation

Ewout van den Berg, Sergey B. Bravyi, and Dmitri Maslov
IBM T.J. Watson Research Center, Yorktown Heights, NY 10598, USA
Dec 2021

Model-free readout-error mitigation for quantum expectation values

Ewout van den Berg, Zlatko K. Minev, and Kristan Temme
Phys. Rev. A **105**, 032620 – Published 30 March 2022

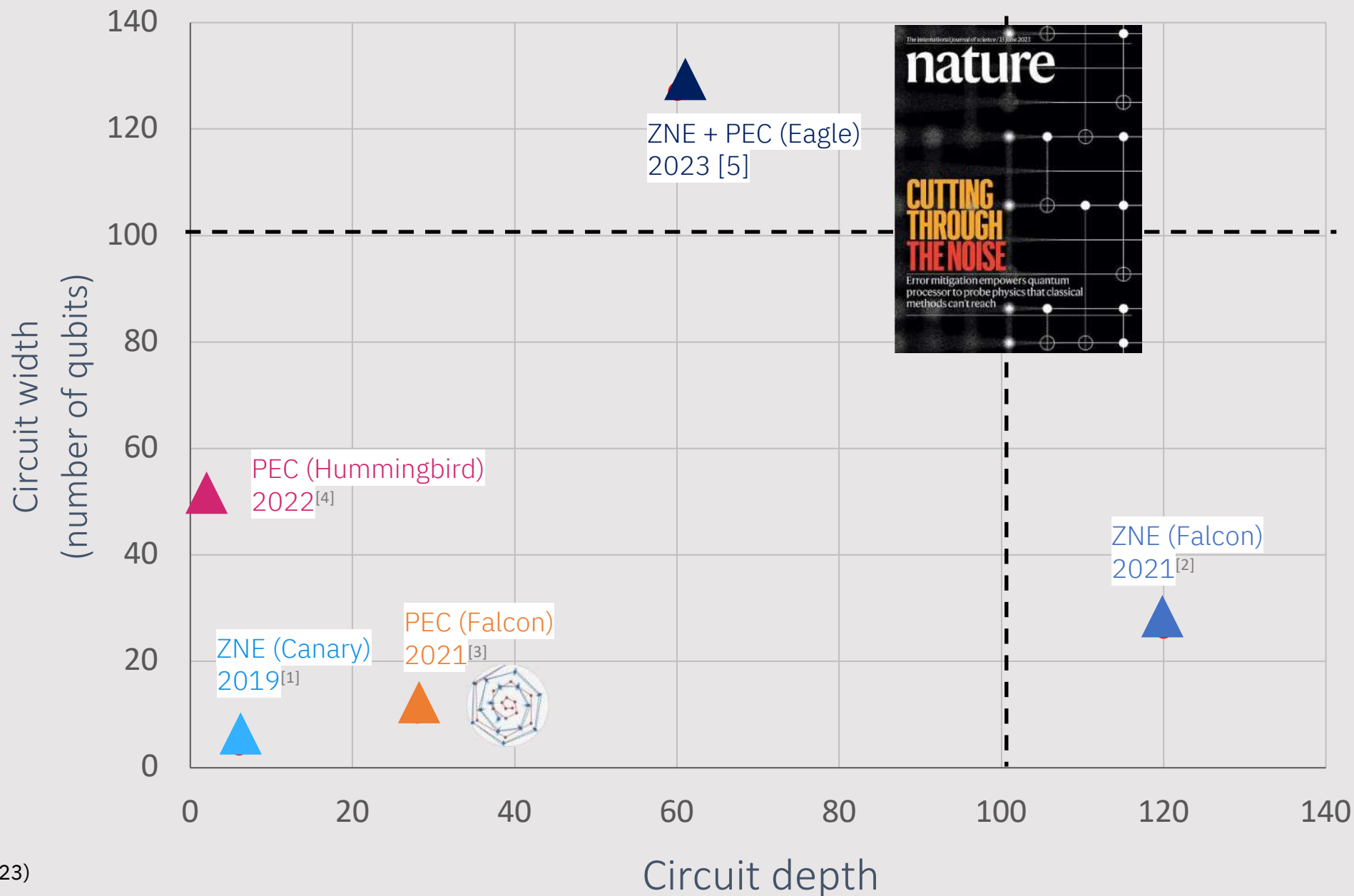
Overview of some key experimental progress in error mitigation:

Error mitigation

No matter what you do you have to chop it to this graph

PEC: Probabilistic error cancelation

ZNE: Zero-noise extrapolation



[1] Kandala, Nature (2019)

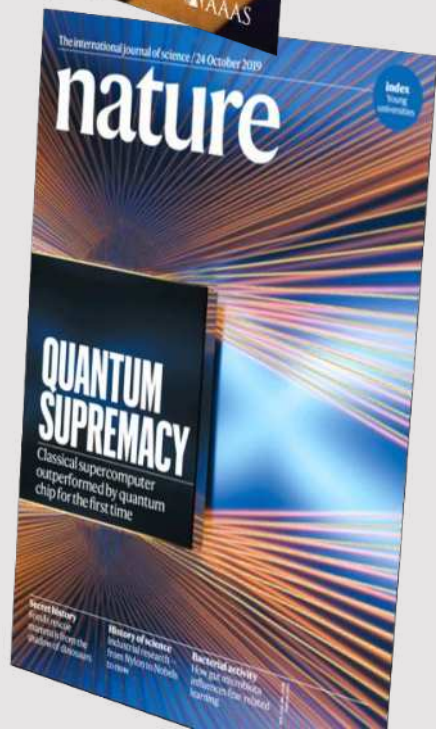
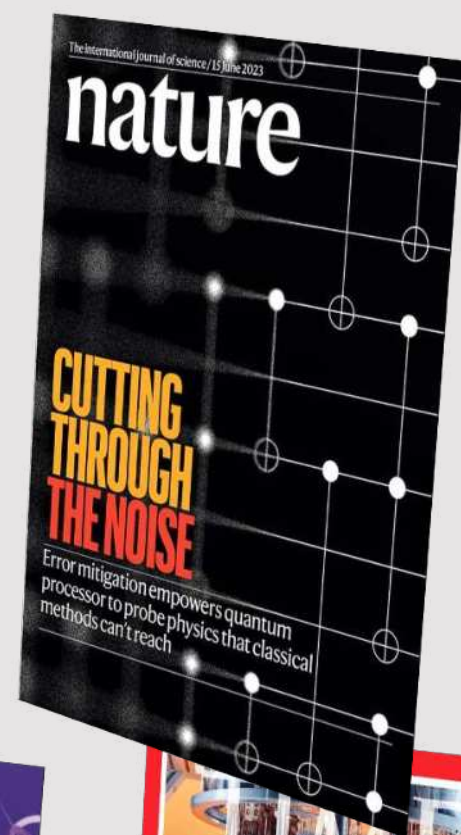
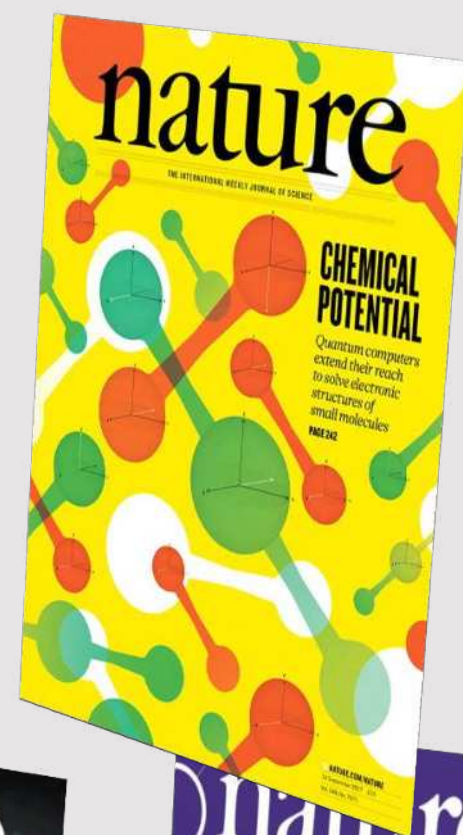
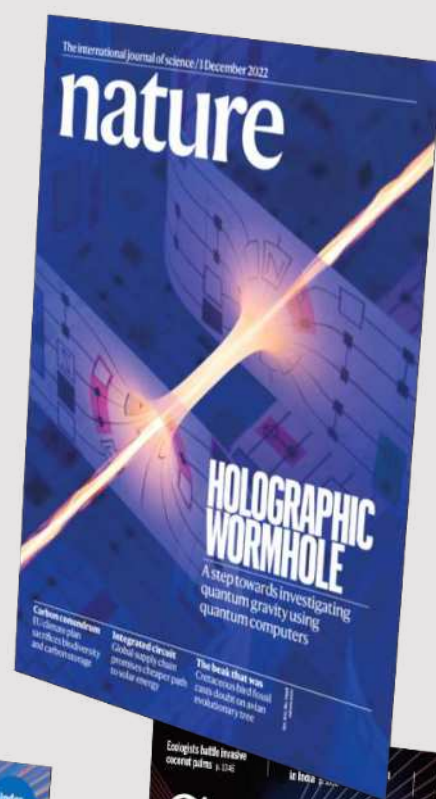
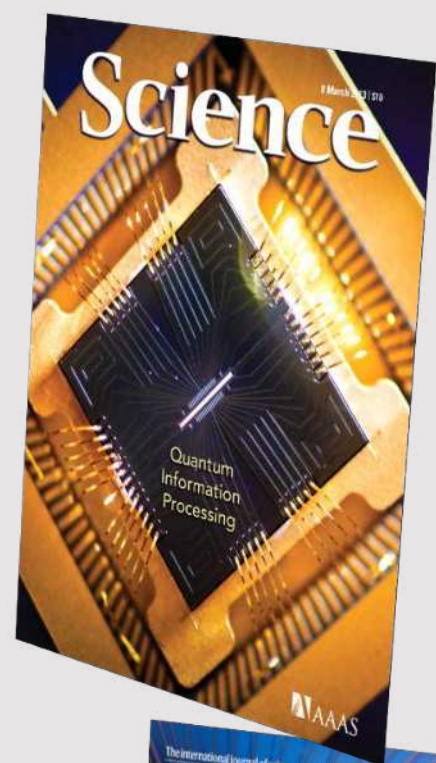
[2] Kim, Nature Phys. (2023)

[3] van den Berg, Mineev, Nature Phys. (2023)

[4] Temme, IBM Research Blog (2022)

[5] Kim, Nature (2023)

Is science with noisy devices of
broad interest today?



Some of these ideas covered in lectures at BSS23, see also BSS seminar by Vedika Khemani

Deep dive:
Probabilistic error cancellation (PEC)
To learn and cancel quantum noise



Got Slides?



Paper: [Nature Physics \(2023\)](#)

Ewout van den Berg, Zlatko K. Mineev, Abhinav Kandala, Kristan Temme

Cancel quantum noise



High-level message

Learn

accurate, efficient, scalable



Cancel

noise with noise,
practical



Cost

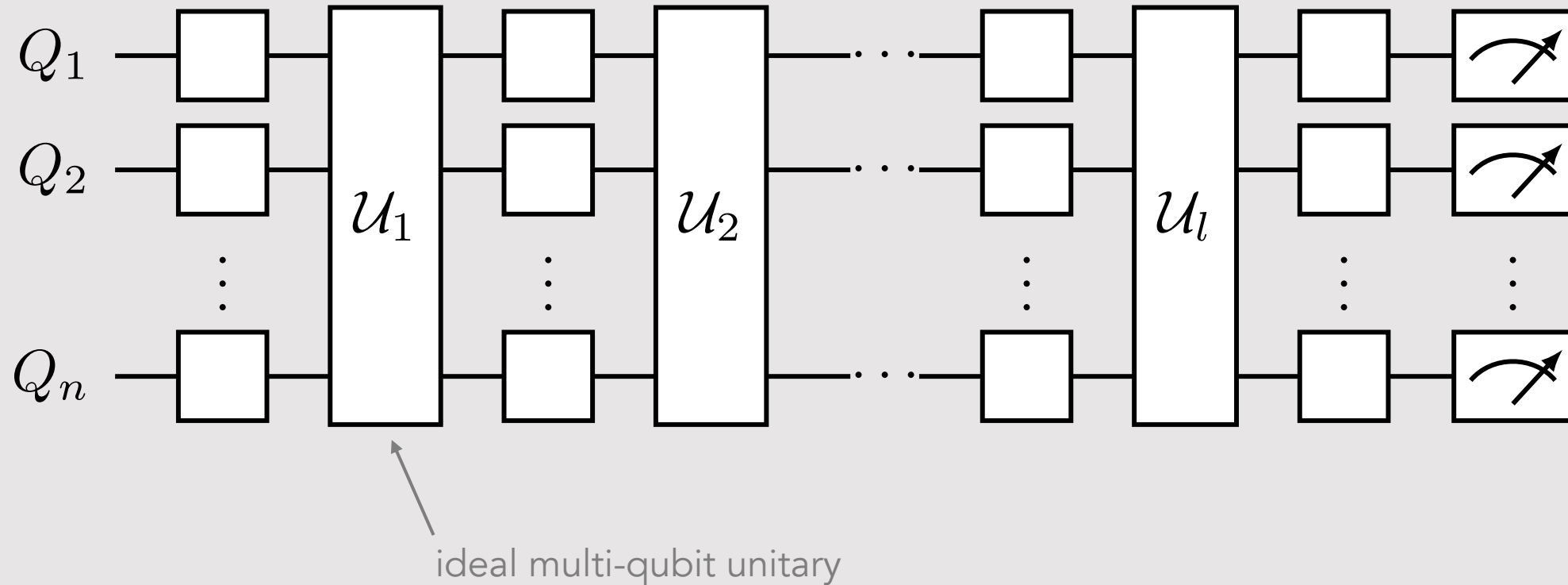
more noise more cost





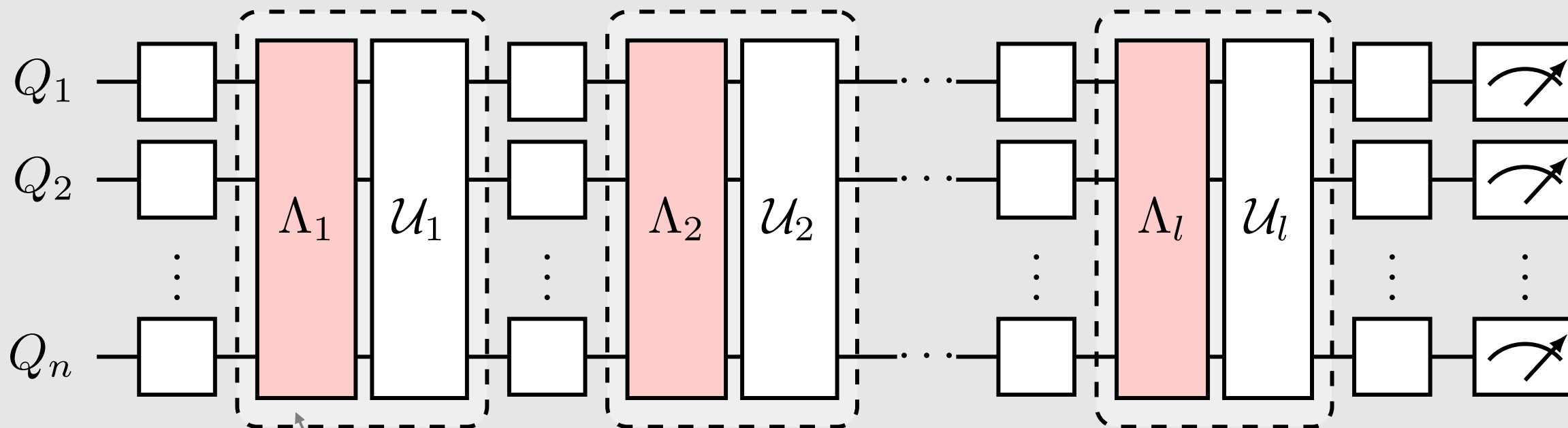
Idea

Ideal (noise-free) quantum circuit



A circuit can be decomposed into a layer construction
Example: Trotterization of Ising model simulation

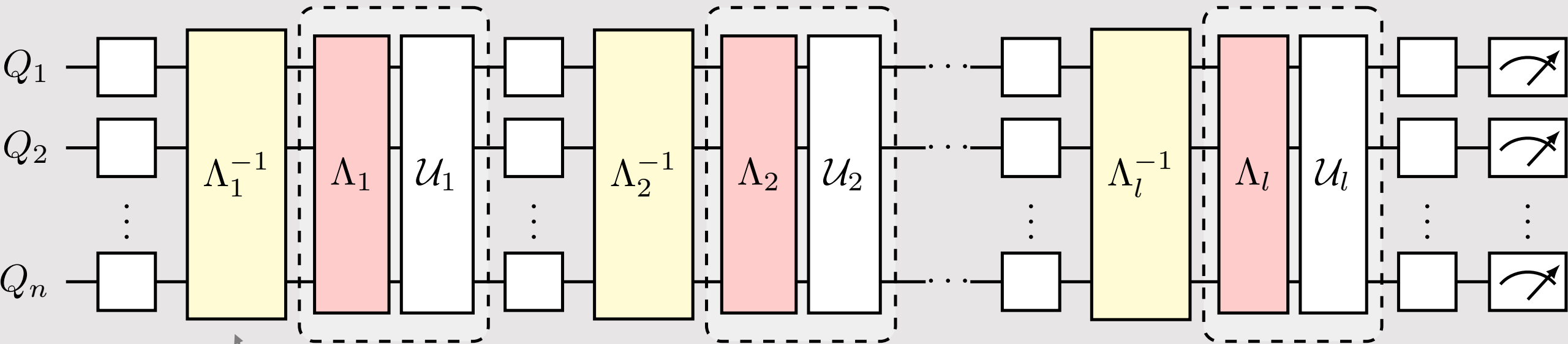
Real (noisy) quantum circuit



multi-qubit noise channel
inseparable from gate

completely positive and trace preserving (CPTP)
representable by a $4^n \times 4^n$ matrix

Why not invert noise?



inverse operation of noise channel

unphysical

would need to know lost information due to noise

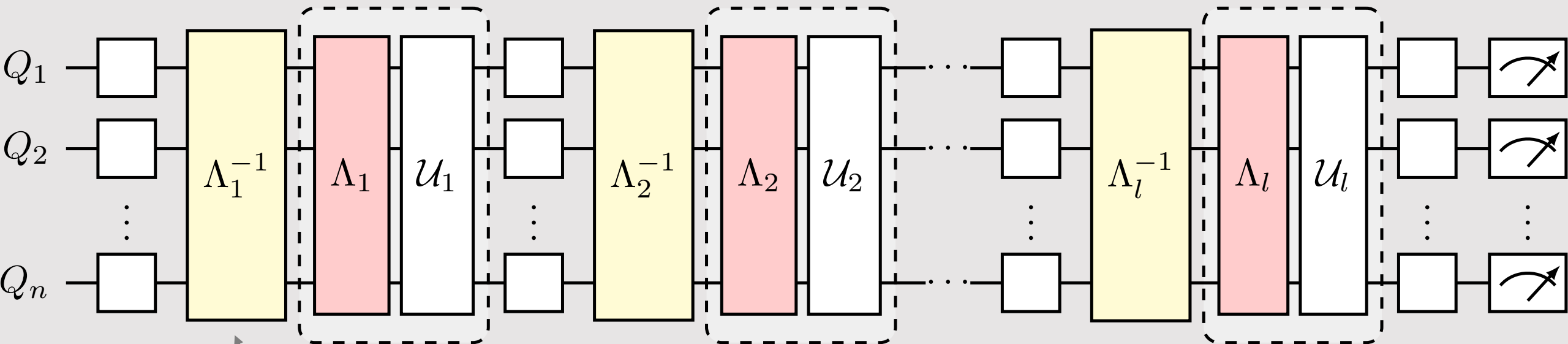
non CPTP map

has negative eigenvalues

...

Not possible?

Probabilistic error cancellation



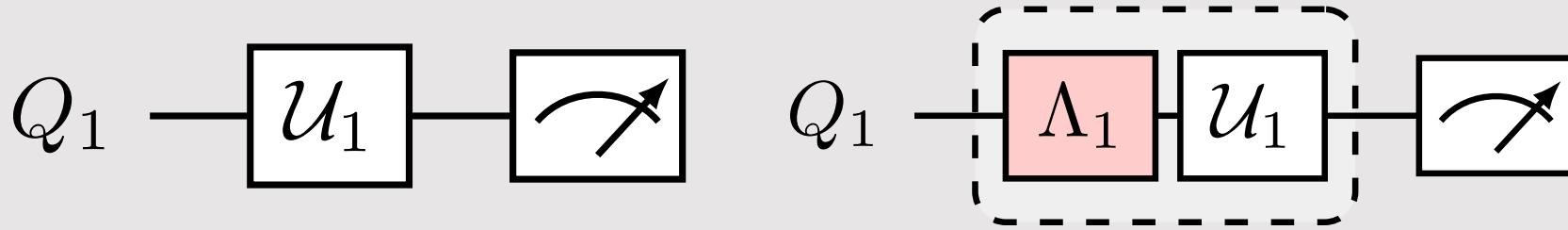
inverse operation of noise channel
implement on average



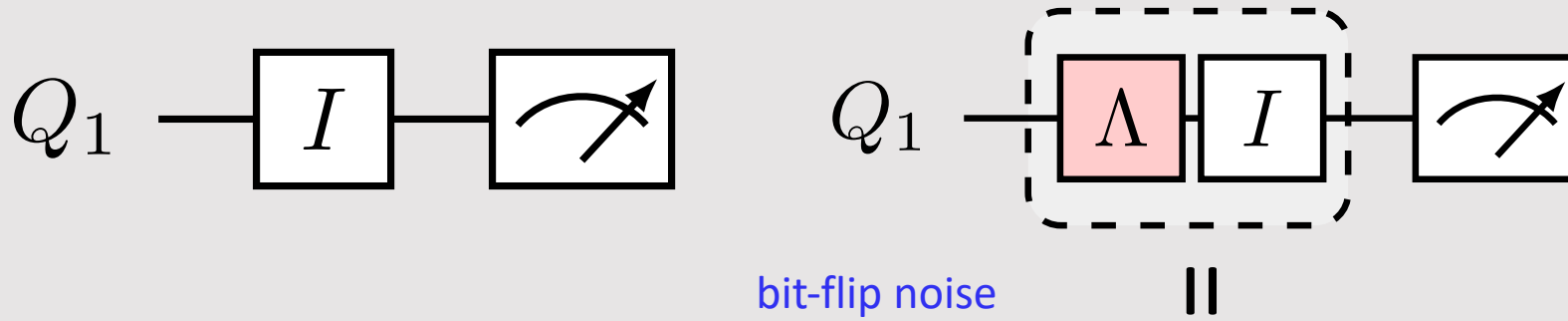
K. Temme, S. Bravyi, and J. M. Gambetta
PRL 119, 180509 (2017)

See also S. Endo, S. Benjamin, and Y. Li
Phys. Rev. X 8, 031027 (2018)

Toy model

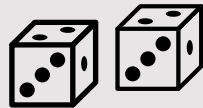


Toy model: noise unraveling into quantum trajectories

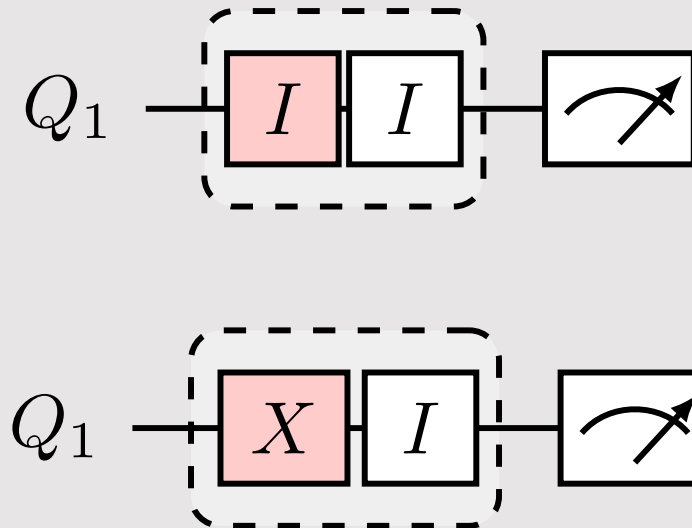


unraveling
(quantum trajectories)

probability $1-p$

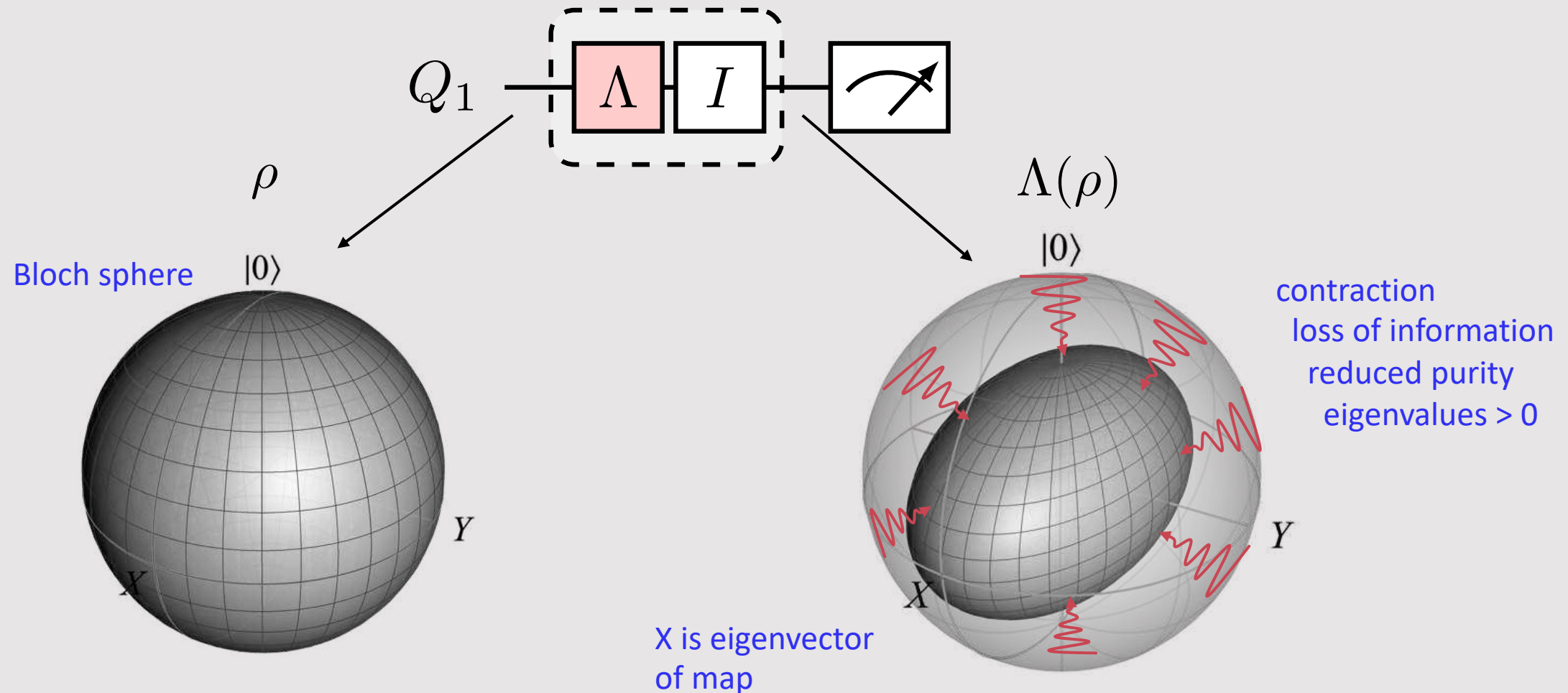


probability p

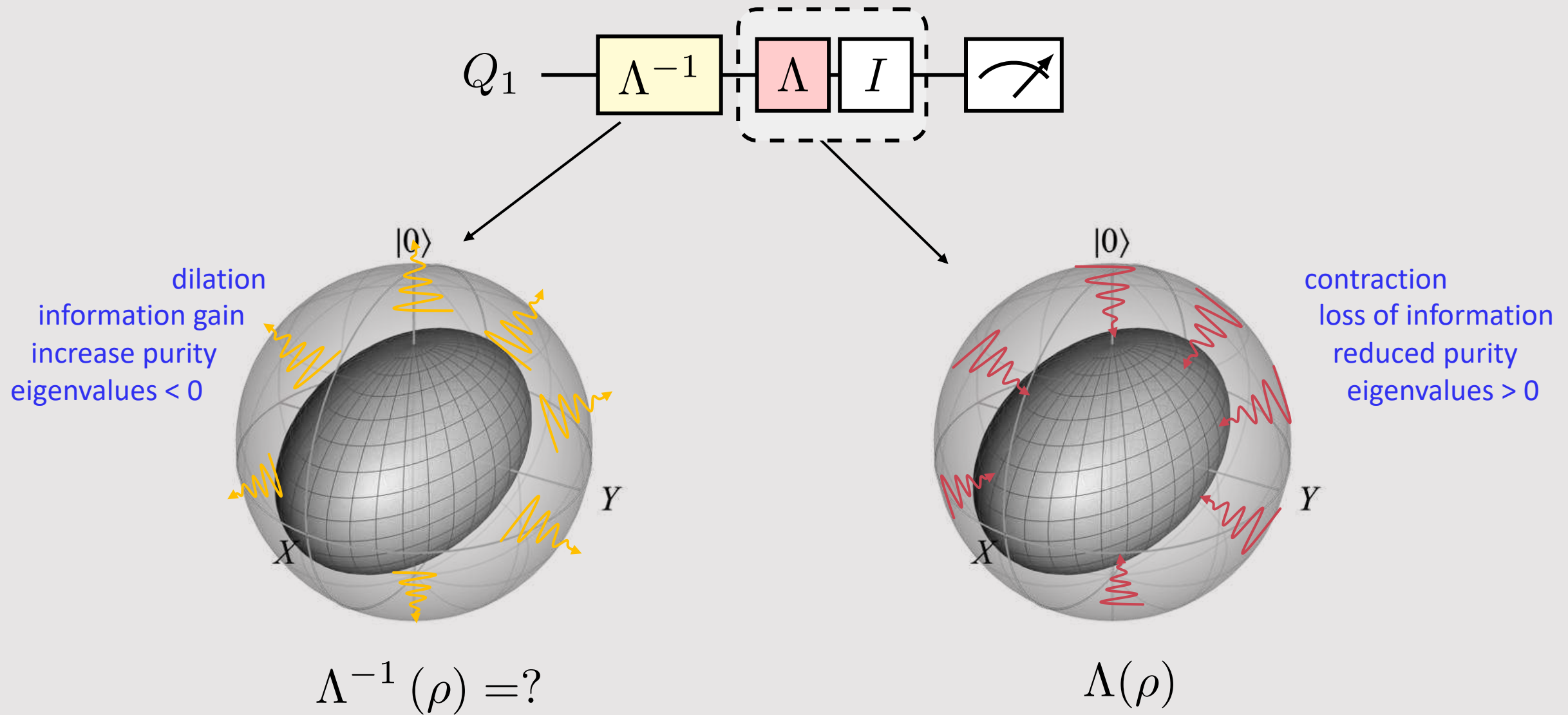


$$\Lambda(\rho) = (1-p)I\rho I + pX\rho X$$

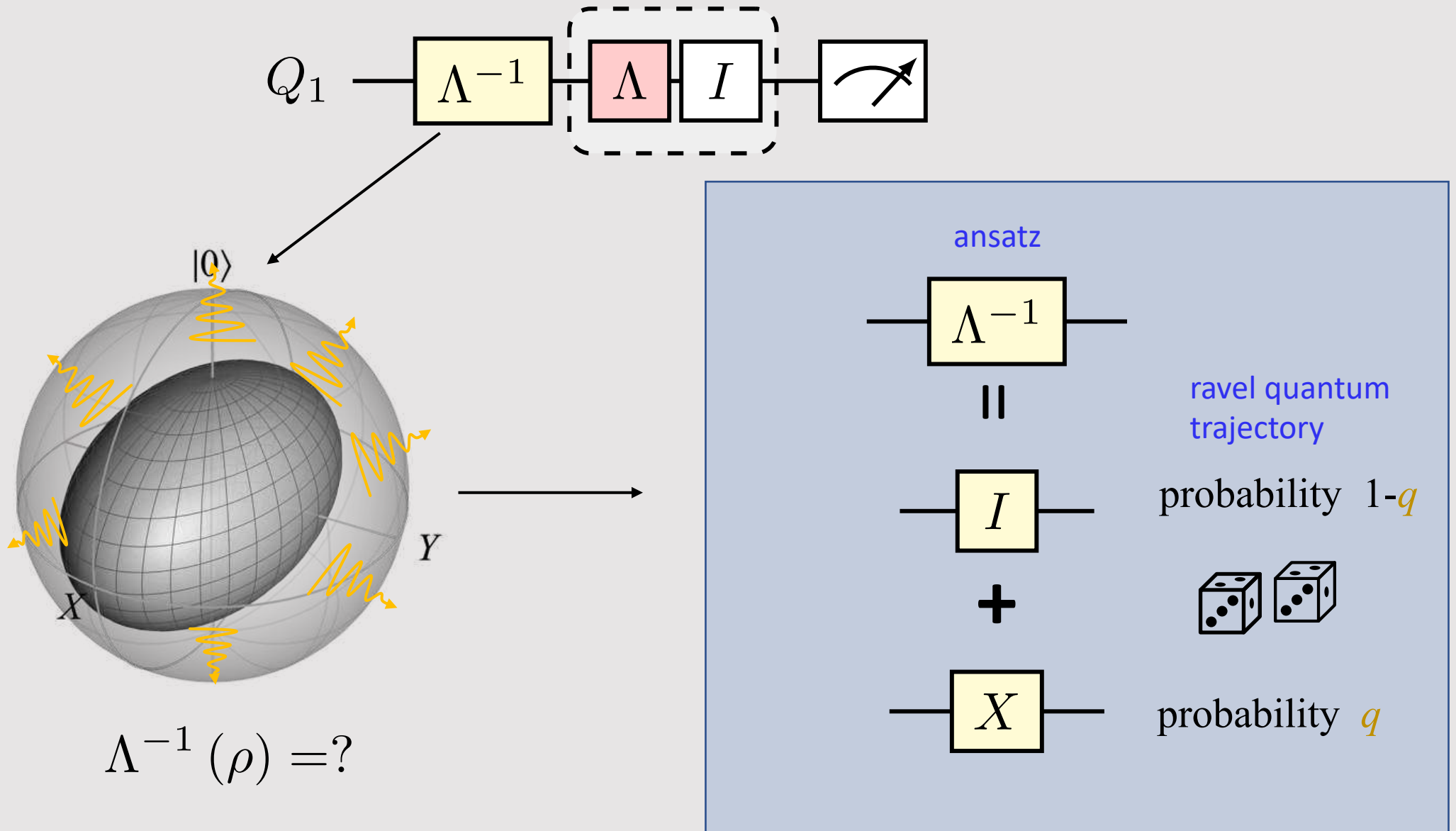
Toy model: noise unraveling into quantum trajectories



Inverse of noise map is not physical

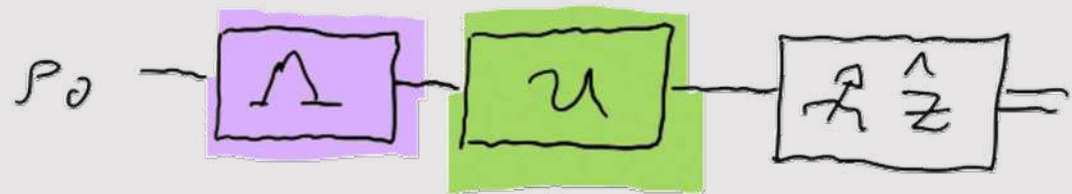
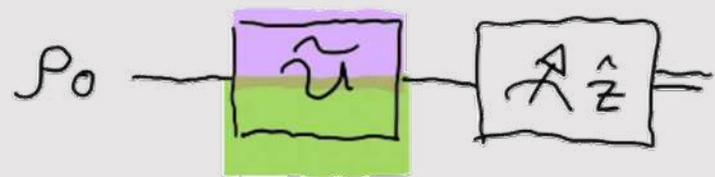
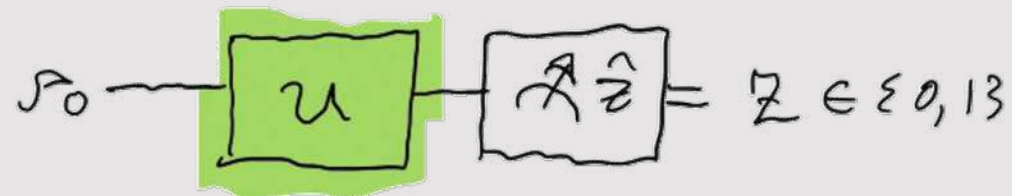


Inverse of noise map is not physical



Blackboard derivation

Setup



Details on notation:

Quantum register alphabet $\Sigma = \{0, 1\}$
Hilbert space $\mathcal{H} = \mathbb{C}^\Sigma$
Initial state $\rho_0 \in D(\mathcal{H}) \subset L(\mathcal{H})$
Ideal unitary $U \in U(\mathcal{H}) \subset L(\mathcal{H})$
Ideal u-channel $\mathcal{U}(\rho) = U \rho U^\dagger$
 $\mathcal{U} \in C(\mathcal{H}) \subset L(L(\mathcal{H}))$

Noisy gate / circuit $\tilde{\mathcal{U}} \in L(L(\mathcal{H}))$

Decompose noisy gate $\tilde{\mathcal{U}} = \mathcal{U} \Lambda$

Blackboard derivation

Simple Example

Keeping it simple and illustrative, let's do a simple case

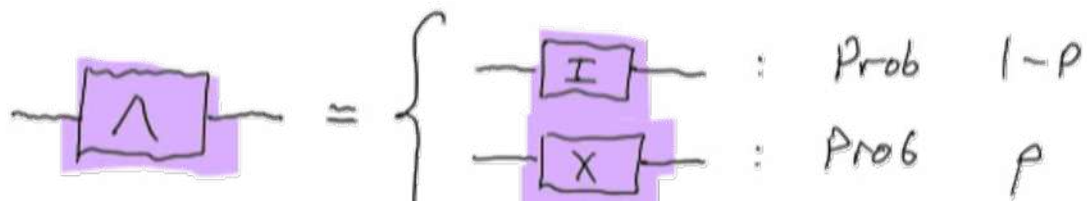
$$\text{Let } U = I \\ u = I \cdot I$$

For the noise, let's play with the simplest bit-flip channel

$$\Lambda(\rho) = \underbrace{(1-p)I\rho I}_{\text{prob of no error}} + \underbrace{pX\rho X}_{\text{prob of a bit flip error}}$$

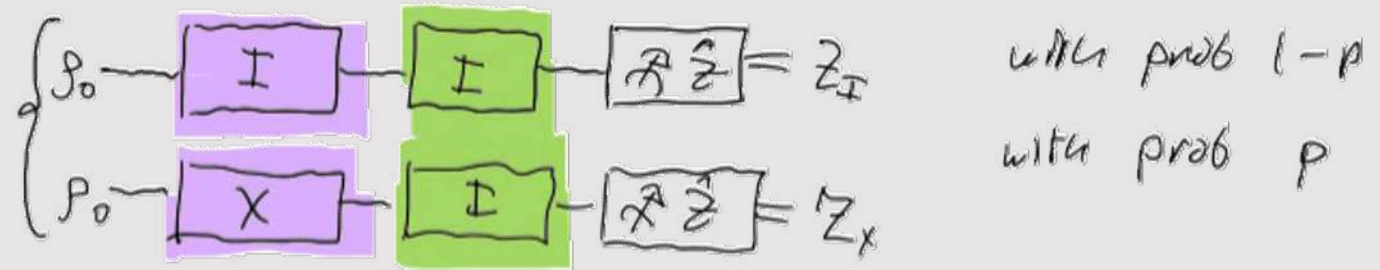
$$\left(\Lambda_\rho = (1-p)\mathcal{I}_\rho + p\mathcal{X}_\rho \right. \quad \left. \begin{array}{l} \text{Equivalent superoperator channel representation} \\ \mathcal{X}_\rho = X\rho X \\ \mathcal{I}_\rho = I\rho I = \rho \end{array} \right)$$

Equivalent trajectory unraveling



Blackboard derivation

Our circuit then is equivalent to either



Simple Example

Keeping it simple and illustrative, let's do a simple case

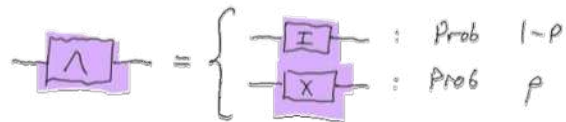
$$\text{Let } U = I \\ \mathcal{U} = I \cdot I$$

For the noise, let's play with the simplest bit-flip channel

$$\mathcal{N}_p = \underbrace{(1-p)I\rho I}_{\text{prob of no error}} + \underbrace{pX\rho X}_{\text{prob of a bit flip error}}$$

$$\left(\begin{aligned} \mathcal{N}_p &= (1-p)\mathcal{I}_p + p\mathcal{X}_p \\ \mathcal{I}_p &= X_p X \\ \mathcal{X}_p &= I \cdot I = p \end{aligned} \right. \quad \begin{array}{l} \text{Equivalent superoperator} \\ \text{channel representation} \end{array}$$

Equivalent trajectory unraveling



The ideal expectation value is

$$\hat{Z}_{\text{ideal}} = \langle \hat{Z} \rangle = \text{Tr}(\hat{Z} \mathcal{I}_p \rho_0) = \text{Tr}(\hat{Z} \rho_0) = \rho_Z$$

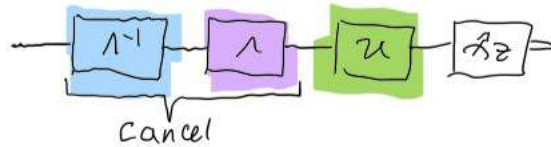
When the channel introduces an error however,

$$\begin{aligned} \mathbb{E}[Z_X] &= \text{Tr}(\hat{Z} X_p X) = \text{Tr}(X \hat{Z} X_p) \\ &= \text{Tr}(-\hat{Z} \rho) \\ &= -\rho_Z \end{aligned}$$

Blackboard derivation

Noise Inverse

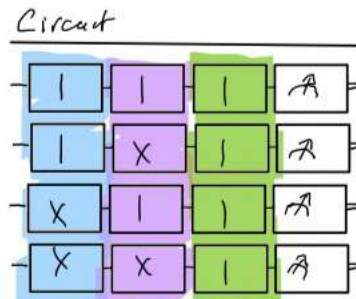
To undo the noise, we'd like to introduce the inverse noise



$$\Lambda^{-1}\Lambda = \Lambda\Lambda^{-1} = \mathbb{I}$$

Taking the ansatz $\Lambda^{-1}(p) = (1-r)\mathbb{I} + r(X \cdot X)$
we see 4 cases of unravelling

inverse	noise	no error	prob
I	I	✓	$(1-r)(1-p)$
I	X	✗	$(1-r)p$
X	I	✗	$r(1-p)$
X	X	✓	rp



ideally, we want to interfere trajectories so that the no-error ones will coherently add to unity probably, and the ones with an error will cancel.

$$\therefore \textcircled{A} \begin{aligned} (1-r)(1-p) + r \cdot p &= 1 \\ 1 - r - p + 2rp &\approx 1 \\ r + p - 2rp &= 0 \end{aligned} \quad \wedge \quad \textcircled{B} \begin{aligned} (1-r)p + r(1-p) &= 0 \\ p + r - 2rp &= 0 \\ \text{same condition} \end{aligned}$$

$$\Rightarrow r(1-2p) = -p$$

$\therefore$

$$r = \frac{-p}{1-2p}$$

Recall p is a probability $0 \leq p \leq 1$,

$$p=0 \Rightarrow r=0$$

$$p=1 \Rightarrow r=1$$

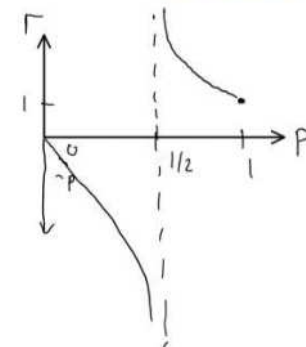
$$p=1/2 \Rightarrow r=\infty$$

$$p \ll 1 \Rightarrow r \approx -p$$

no noise, no need to do anything

deterministic bit-flip, requires deterministic bit flip usually for Λ^{-1}

singular value, since at $p=1/2$, we hit 0, scramble the state



Blackboard derivation

Note that we could equivalently have used the algebraic condition and solved for r

$$\Lambda(\Lambda^{-1}(\rho)) = \mathcal{I}(\rho) = \rho \quad \text{Solve for } r$$

$$= \Lambda((1-r)\rho + rX\rho X)$$

$$= \underbrace{(1-p)(1-r)\rho + prX\rho X}_{\text{no error}} + \underbrace{(1-p)rX\rho X + (1-r)pX\rho X}_{\text{error}}$$

$$= \underbrace{[(1-p)(1-r) + pr]\rho}_{\text{no error}} + \underbrace{[(1-p)r + (1-r)p]X\rho X}_{\text{error}}$$

Same conditions as above! solution $r = \frac{p}{1-2p}$

Blackboard derivation

How to implement? Quasi-Probability

$$\begin{aligned}\Lambda^{-1} &= (1-r)I_P I + r X_P X \\ &= \left[\frac{|1-r|}{|1-r|+|r|} \operatorname{sgn}(1-r) I_P I + \frac{|r|}{|1-r|+|r|} \operatorname{sgn}(r) X_P X \right] (|1-r|+|r|) \\ &= \gamma \left[S_I P_I I_P I + S_X P_X X_P X \right]\end{aligned}$$

with

$$\gamma = |1-r| + |r|$$

$$P_I = \frac{|1-r|}{\gamma} \quad S_I = \operatorname{sgn}(1-r)$$

$$P_X = \frac{|r|}{\gamma} \quad S_X = \operatorname{sgn}(r)$$

valid prob distribution

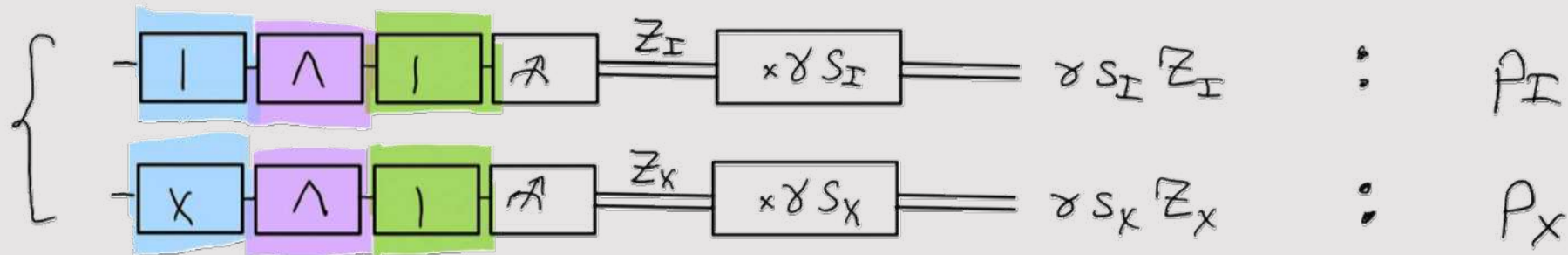
$$0 \leq P_I, P_X \leq 1 \quad \text{and} \quad |P_I| + |P_X| = 1$$

Blackboard

How to sample?

$$\begin{aligned}
 \langle Z \rangle &= \text{Tr}(Z \Lambda \Lambda^{-1} \rho_0) \\
 &= \text{Tr}(Z \tilde{I} [\gamma S_I \rho_I \rho_0^\dagger + \gamma S_X \rho_X \rho_0^\dagger X]) \\
 &= \gamma S_I \rho_I \text{Tr}(Z \tilde{I} \rho_0) + \gamma S_X \rho_X \text{Tr}(Z \tilde{I} X \rho_0) \\
 &= \gamma \left[S_I \rho_I \underbrace{\langle Z \rangle_I}_{\substack{\text{quantum} \\ \text{circuit exp. val use can find}}} + S_X \rho_X \underbrace{\langle Z \rangle_X}_{\leftarrow} \right]
 \end{aligned}$$

Equivalent interpretation:



Blackboard

Estimator

$$E_{\text{mitg}} = \gamma S_I Z_I + \gamma S_X Z_X$$

$$\mathbb{E}[E_{\text{mitg}}] = \langle \hat{Z} \rangle_{\text{ideal}}$$

$$\begin{aligned} \mathbb{V}[E_{\text{mitg}}] &= \mathbb{V}[\gamma S_I Z_I] + \mathbb{V}[\gamma S_X Z_X] \\ &= \gamma^2 \mathbb{V}[Z_I] + \gamma^2 \mathbb{V}[Z_X] \\ &= \gamma^2 (2 \sigma_{\text{ideal}}^2) \end{aligned}$$

$$\sigma_{\text{ideal}}^2 = \mathbb{V}[Z_I] = 4q(1-q)$$

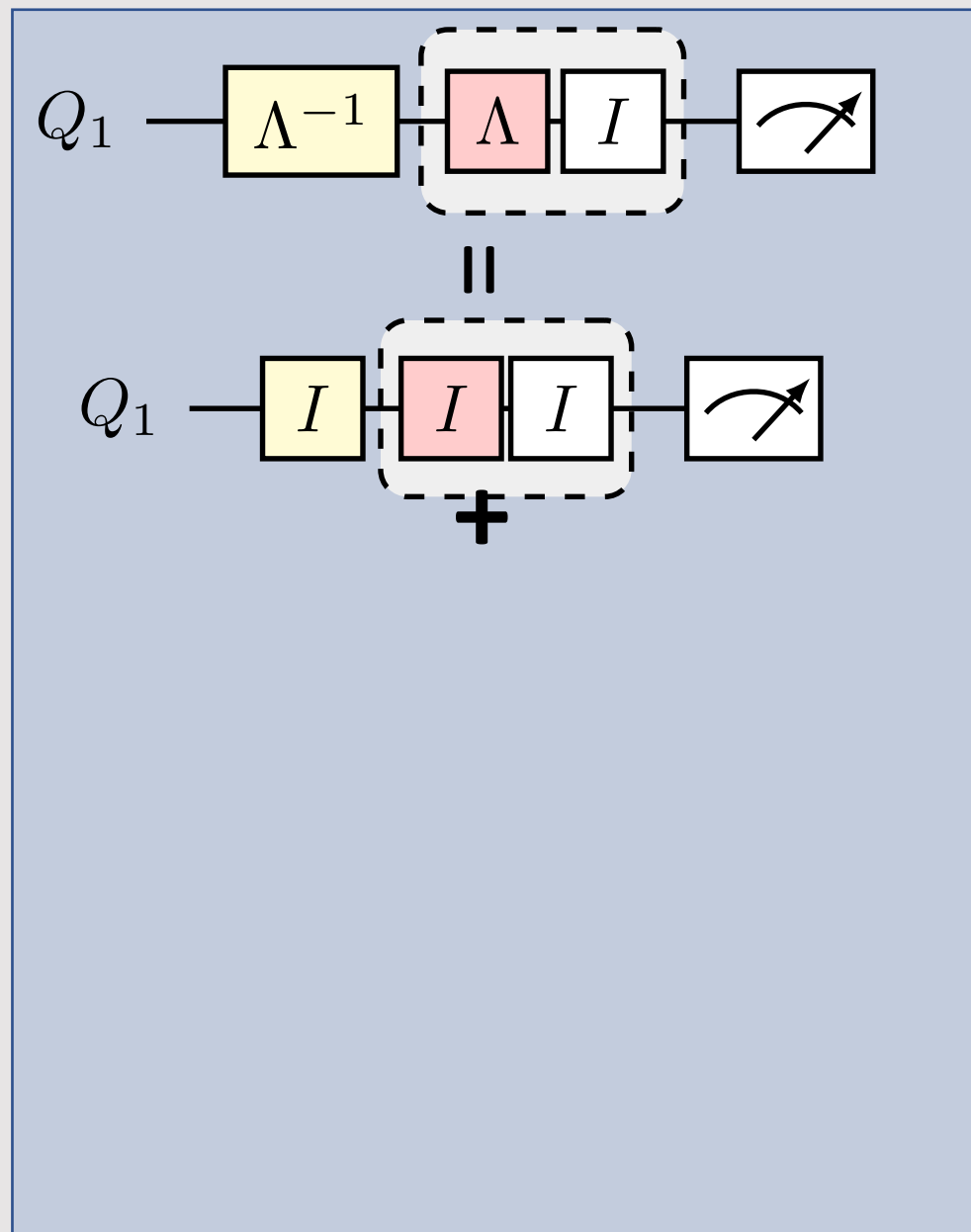
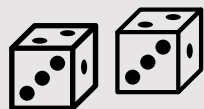
$$q = 1 - 2P_Z = \langle \frac{1 - \hat{Z}}{2} \rangle$$

Since the X just flips $Z \rightarrow -Z$ of P , it follows that the variance is the same, since symmetric

Raveling quantum trajectories to undo noise

No error

probability
 $(1-q)(1-p)$



Raveling quantum trajectories to undo noise

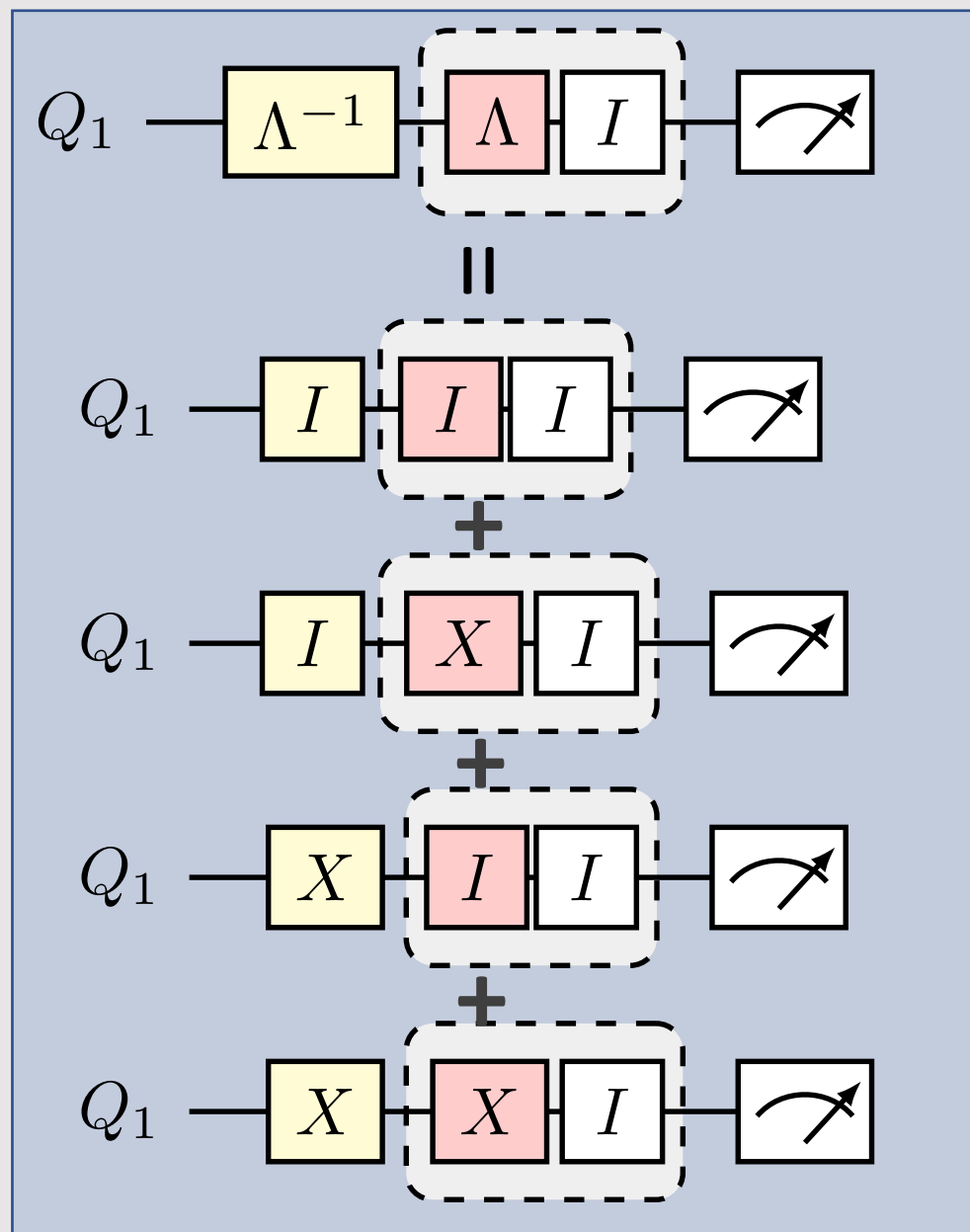
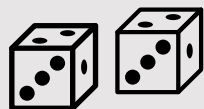
probability

No error $(1-q)(1-p)$

ERROR! $(1-q)p$

ERROR! $q(1-p)$

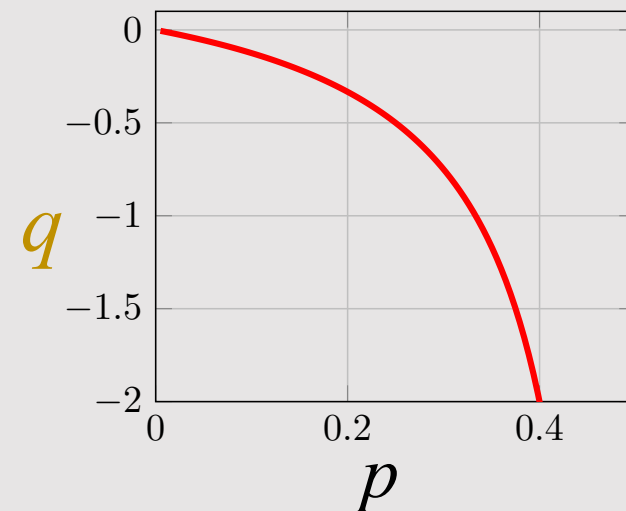
Error CANCELED! qp



Solution to noise free!

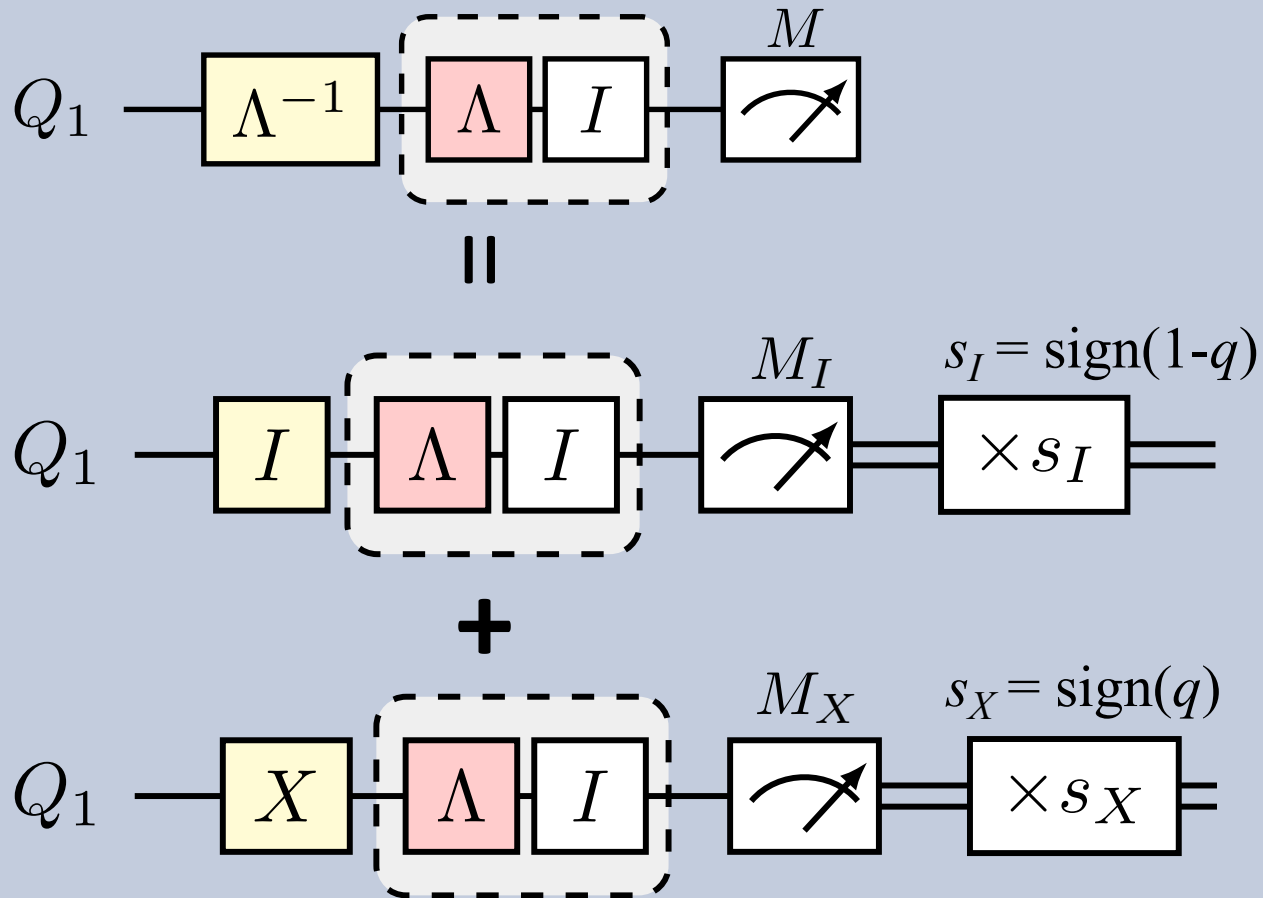
$$q = \frac{-p}{1 - 2p}$$

Sign & scale:
quasi-probability



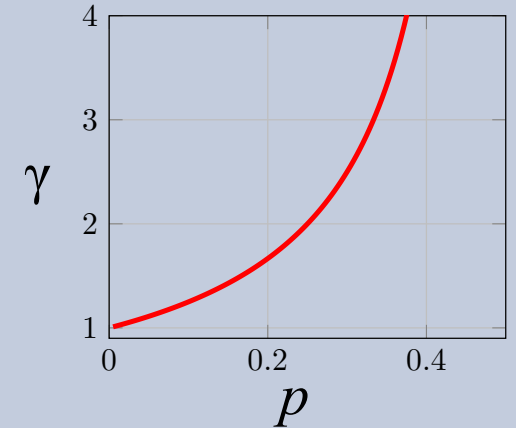
How to implement?

random circuits
(quantum trajectories)



sampling overhead

$$\gamma = |1-q| + |q|$$



mitigated expectation

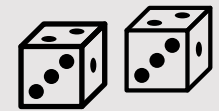
$$\langle M \rangle = \gamma (s_I P_I M_I + s_X P_X M_X)$$

Gain: Bias-free estimate!

Cost: Variance

probability

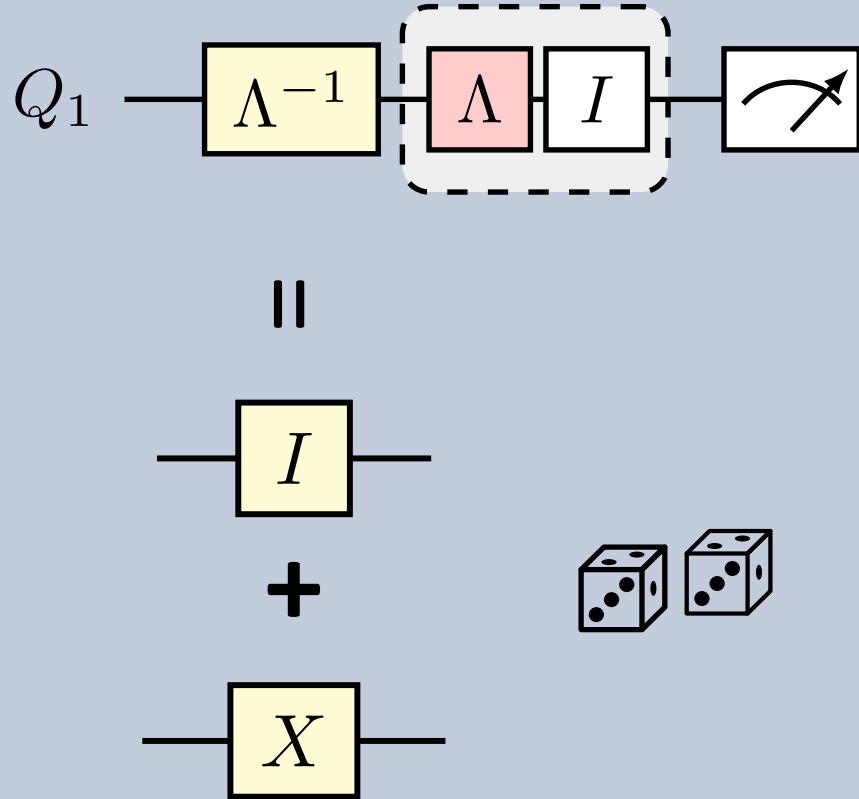
$$P_I = |1-q|/\gamma$$



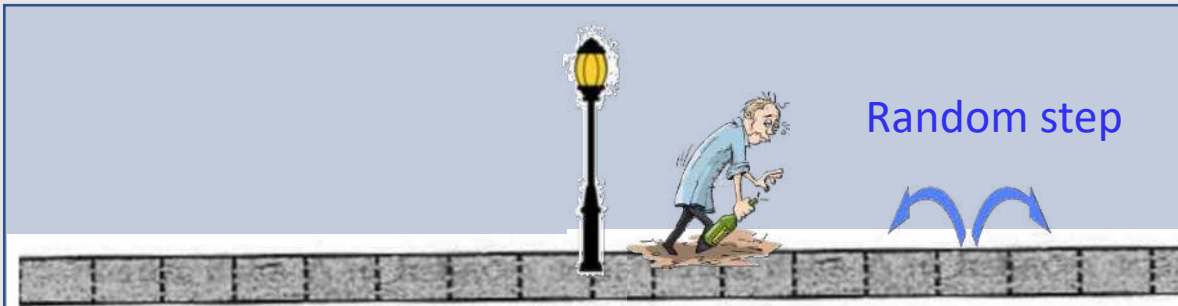
$$P_X = |q|/\gamma$$

$$P_I + P_X = 1$$

Canceling noise with noise

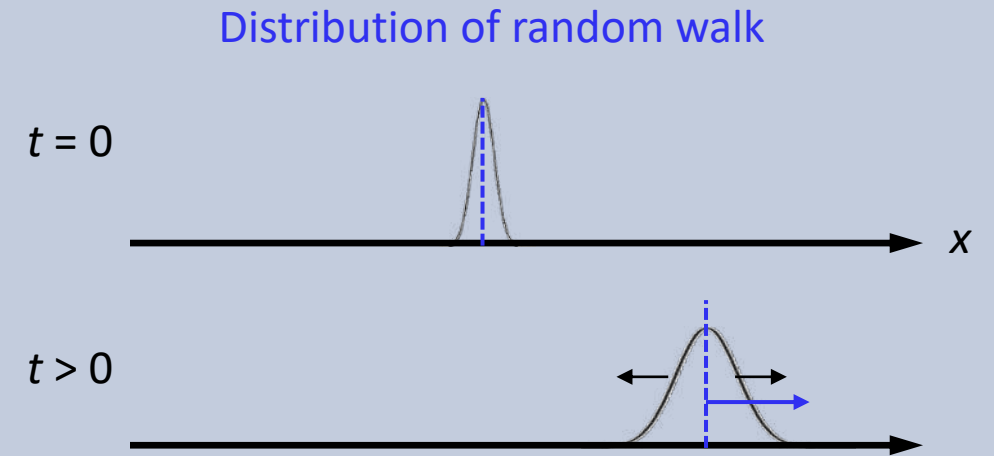


Canceling noise with noise: Drunkard's classical random walk analogy



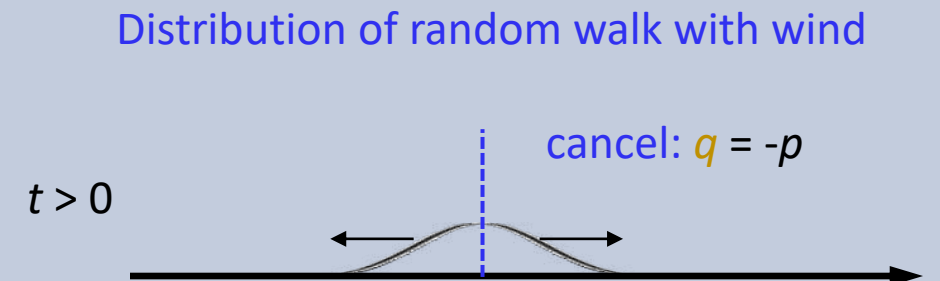
$$P(1 \text{ step left}) = \frac{1}{2} - p$$

$$P(1 \text{ step right}) = \frac{1}{2} + p$$



$$P(1 \text{ step left}) = \frac{1}{2} + q$$

$$P(1 \text{ step right}) = \frac{1}{2} - q$$



Gain: Bias-free estimate!

Cost: Variance